Artificial Intelligence Driven Meta-Analysis

Introduction to Meta Analysis



Dr. Wen-Wang Rao （饶文旺）

Lecturer

E-mail: raoww@mpu.edu.mo

Faculty of Applied Sciences

Macao Polytechnic University





Reminder for Group Project

1. **Date of Final Report：** Dec. 5, 2025
   - **Deadline of Submission：** 10.00 am
2. **Date of Presentation：** Dec. 12, 2025
   - **Time：** 10:00-13:00 and 19:00-22:00
   - **Deadline of Submission：** 10.00 am



E-mail: Dr. Rao: raoww@mpu.edu.mo

Major Requirements for Group Project

- Teams consist of **3 or 4 randomly** assigned members
- The project data will be released on **Sep 12, 2026** (today)
- Your team need to select **a topic** and utilize it to develop a detailed project.
- Teams are required to submit ***the final project reports with a Turnitin report, the ppt or pdf of the presentation slides, the work contribution and Gen AI documents***.
- Teams should select their **own presenter**.
- The project leader will submit **all project files**, and every member must submit a **peer evaluation for the entire team.**
- **For more details, please refer to the files in Canvas.**



Systematic Review And Meta-Analysis

A Step by Step guide Approach


Garbage in
Garbage OUT

Determinants of quality of evidence (2)

Quality	Study design	Lower quality if:	Higher quality if:
High (++++)	Randomised controlled trials	Risk of bias (-1, -2) Inconsistency (-1, -2) Indirectness (-1, -2) Imprecision (-1, -2) Publication bias (-1, -2)	Large effect (+1, +2) Dose-response (+1) Direction of residual confounding and biases (+1)
Moderate (+++)			
Low (++)	Observational studies		
Very low (+)	Case reports, expert opinion, modelling		

Research Design: Methodological Planning

- In randomized controlled trials, the highest quality level is the highest, therefore the level is high;
- In observational studies (including cohort studies, case-control studies, and cross-sectional studies), the quality level is low;
- For case reports, case series, and other study designs, the quality level is extremely low.

⊕⊕⊕⊕/A

⊕⊕oo/C

⊕ooo/D

5

- High risk of bias
- Inconsistency
- Indirectness
- Imprecision
- Publication bias
- Reporting bias

3

- Large effect size
- Dose-response gradient
- Residual confounding
- Plausible confounding or bias that would reduce an effect

Grading strength of a recommendation

GRADE strength of recommendation

Definition: The degree of confidence that desirable effects of adherence to a recommendation outweigh undesirable effects.

- **Strong** recommendation:
  The panel is confident that the desirable effects of adherence to a recommendation outweigh the undesirable effects.

- **Weak/ conditional/ discretionary** recommendation:
  The panel concludes that the desirable effects of adherence to a recommendation probably outweigh the undesirable effects, but is not confident.

Low quality of evidence, strong recommendation?

- Clear distinction between
  - quality of evidence, driven by confidence in body of evidence
  - strength of recommendation, driven by confidence in body of evidence plus other considerations
- All combinations possible, e.g.:
  - High quality of evidence -> weak recommendation
  - Very low quality of evidence -> strong recommendation
- Paradigmatic situations for strong recommendations in the absence of high-quality evidence (Alexander et al. 2016)

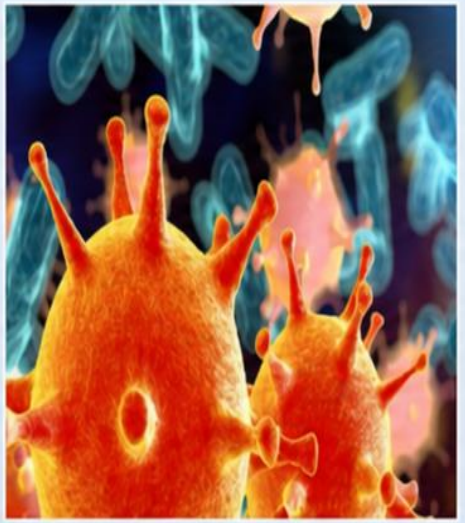
Combined Therapy (Integrated Treatment) VS **Monotherapy**

Actual use depends on the physician's or patient's choice.



High-quality Evidence → Weak Recommendation Strength

Severe Infectious Disease Treatment Options



Benefits of Antimicrobial Therapy



Low/Very Low Quality Evidence → Strong Recommendation

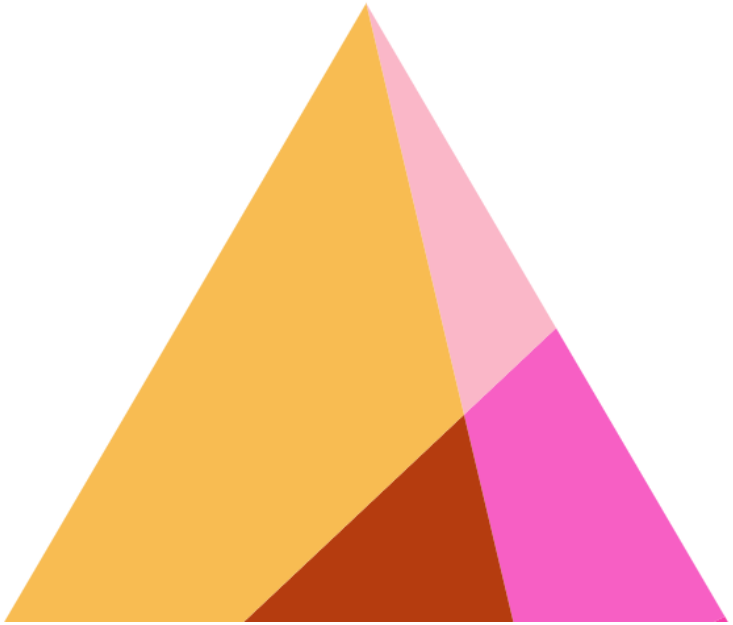
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Join us on the last GRADEpro Webinar of 2022!
December 12, 2022 at 17:00 - 18:00 CET (08:00 - 09:00 PST, 11:00 - 12:00 EST)

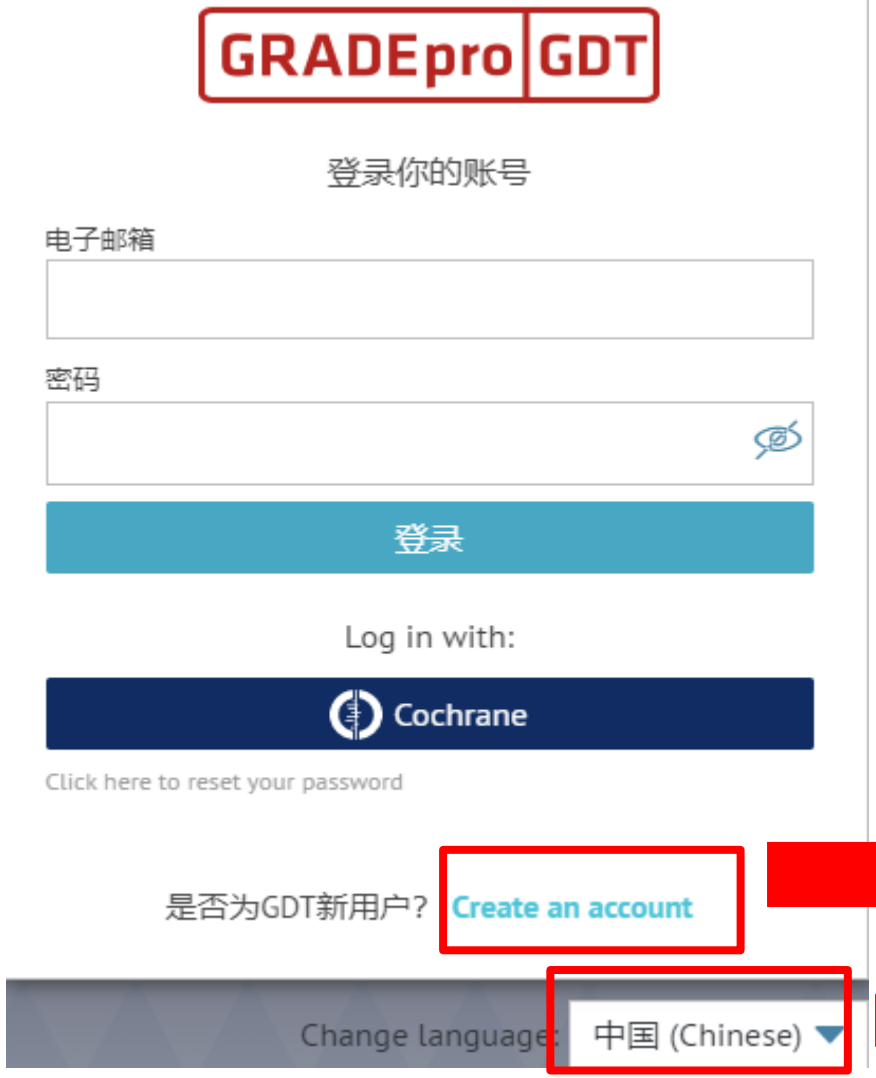
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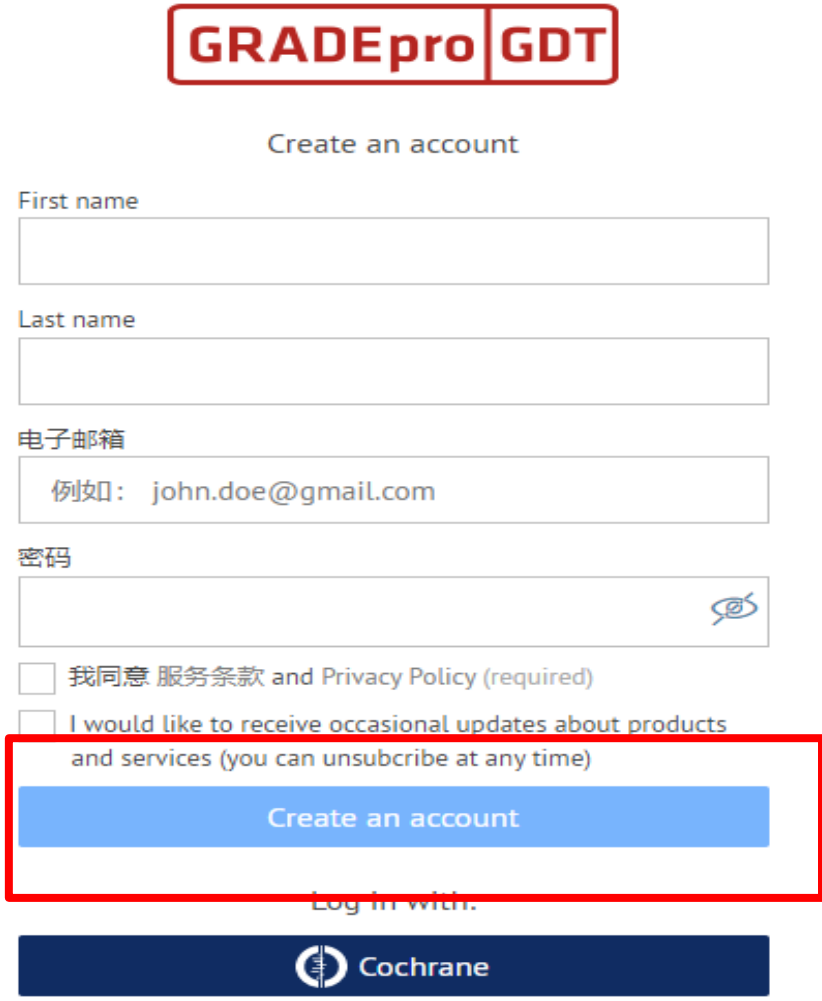
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Create an account
First name
Last name
电子邮箱
例如： john.doe@gmail.com
密码
我同意 服务条款 and Privacy Policy (required)
I would like to receive occasional updates about products and services (you can unsubcribe at any time)
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Research question

- **Does exercise therapy improve symptom remission in adults with depression?**

PICO

- **P—Population:** Adults with depression
- **I—Intervention:** Structured exercise therapy
- **C—Comparator:** Usual care
- **O—Outcome:** Symptom remission at 12 weeks

Randomized controlled trials generally starts at **high certainty**



Included studies

The systematic review included:

- Eight randomized controlled trials;
- 1,200 participants;
- Follow-up periods ranging from 8 to 16 weeks.

The meta-analysis produced the following results:

- Risk ratio: **RR = 1.25**
- 95% confidence interval: **1.05–1.49**
- Heterogeneity: **I^2 = 58%**

If 300 per 1,000 participants achieve symptom remission with usual care, approximately 375 per 1,000 would be expected to achieve remission with exercise therapy. This corresponds to approximately 75 additional cases of remission per 1,000 participants, with a possible range of 15 to 147 additional cases.



Risk of bias

Findings

- Three of the eight studies did not adequately report allocation concealment;
- Blinding participants to an exercise intervention was not feasible;
- Two studies had attrition rates greater than 20%;
- Some studies did not clearly report whether intention-to-treat analysis was performed.

Judgment

There were concerns about study design and conduct that could have affected the results.

Decision

Downgrade by one level for serious risk of bias



Inconsistency

Findings

- The meta-analysis had an I^2 value of 58%, suggesting moderate-to-substantial heterogeneity;
- Most studies reported effects in the same direction, but the magnitude of the effects varied considerably;
- Subgroup analyses did not fully explain the heterogeneity

Judgment

There were important unexplained differences among the study results.

Decision

Downgrade by one level for serious inconsistency



Indirectness

Findings

- The study populations were consistent with the target population of adults with depression;
- The interventions involved structured exercise therapy;
- The comparator was usual care;
- The outcome directly measured symptom remission;
- The follow-up periods were generally appropriate for the review question.

Judgment

The evidence directly addressed the review's PICO question.

Decision

Do not downgrade



Imprecision

Findings

- The total sample size was 1,200 participants;
- The 95% confidence interval ranged from 1.05 to 1.49;
- The confidence interval did not cross the line of no effect, RR = 1.00;
- However, the lower limit of RR = 1.05 was close to no effect.

Judgment

If the prespecified threshold for a clinically important effect was RR = 1.10, the confidence interval would include both a clinically unimportant effect and a substantial benefit. Therefore, the result could be considered imprecise.

For this example, however, assume that the review team concluded that the degree of imprecision was not serious after considering the sample size, number of events, and absolute effect.

Decision

Do not downgrade, but explain the judgment in a footnote.



Publication bias

Findings

- Eight studies were included;
- The funnel plot did not show obvious asymmetry;
- Trial registries and grey literature sources were searched;
- However, the small number of studies made it difficult to rule out publication bias reliably.

Judgment

There was insufficient evidence of serious publication bias.

Decision

Do not downgrade.



Large magnitude of effect

If a study shows a very large effect that cannot reasonably be explained by bias or confounding, the certainty of the evidence may be rated up.

Traditional GRADE guidance provides the following reference thresholds:

- **Large effect:** RR > 2.0 or RR < 0.5—consider rating up by one level;
- **Very large effect:** RR > 5.0 or RR < 0.2—consider rating up by two levels.

Judgment

RR=1.25, 95%CI=1.05–1.49; An RR of 1.25 indicates only a 25% relative increase in remission and is far below the reference threshold of RR > 2.0 for a large effect.

Decision

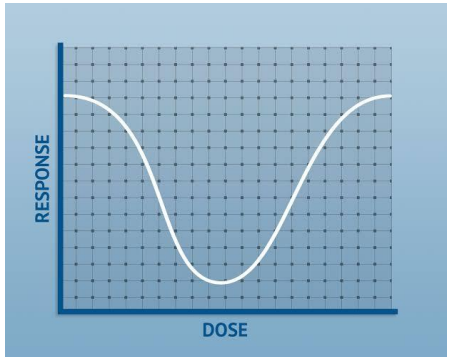
The certainty of the evidence would not be rated up for a large magnitude of effect.

EXAMPLE

Dose–response gradient

If an outcome changes systematically as the intervention dose, exposure level, treatment intensity, or treatment duration increases, this may strengthen confidence in a causal relationship.

Judgment
Structured exercise therapy versus usual care.

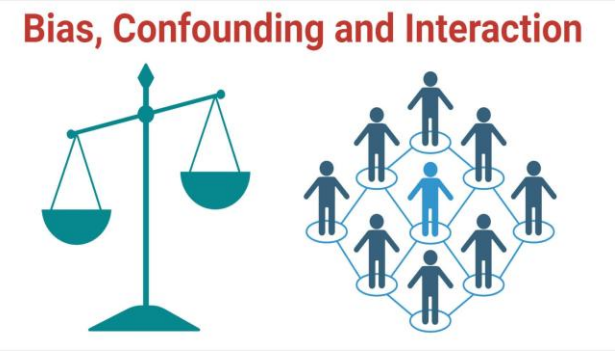


Decision

There is no evidence supporting rating up for a dose–response gradient.

EXAMPLE

All plausible residual confounding or bias would reduce the observed effect



Judgment

There is no such evidence. on the contrary:

- Allocation concealment was unclear;
- Attrition was high in some studies;
- The use of intention-to-treat analysis was unclear;
- The subjective outcome may have been affected by the lack of blinding.

These limitations could exaggerate the effect of exercise therapy rather than necessarily move the estimate toward the line of no effect. The criterion for rating up is therefore not met.

Decision

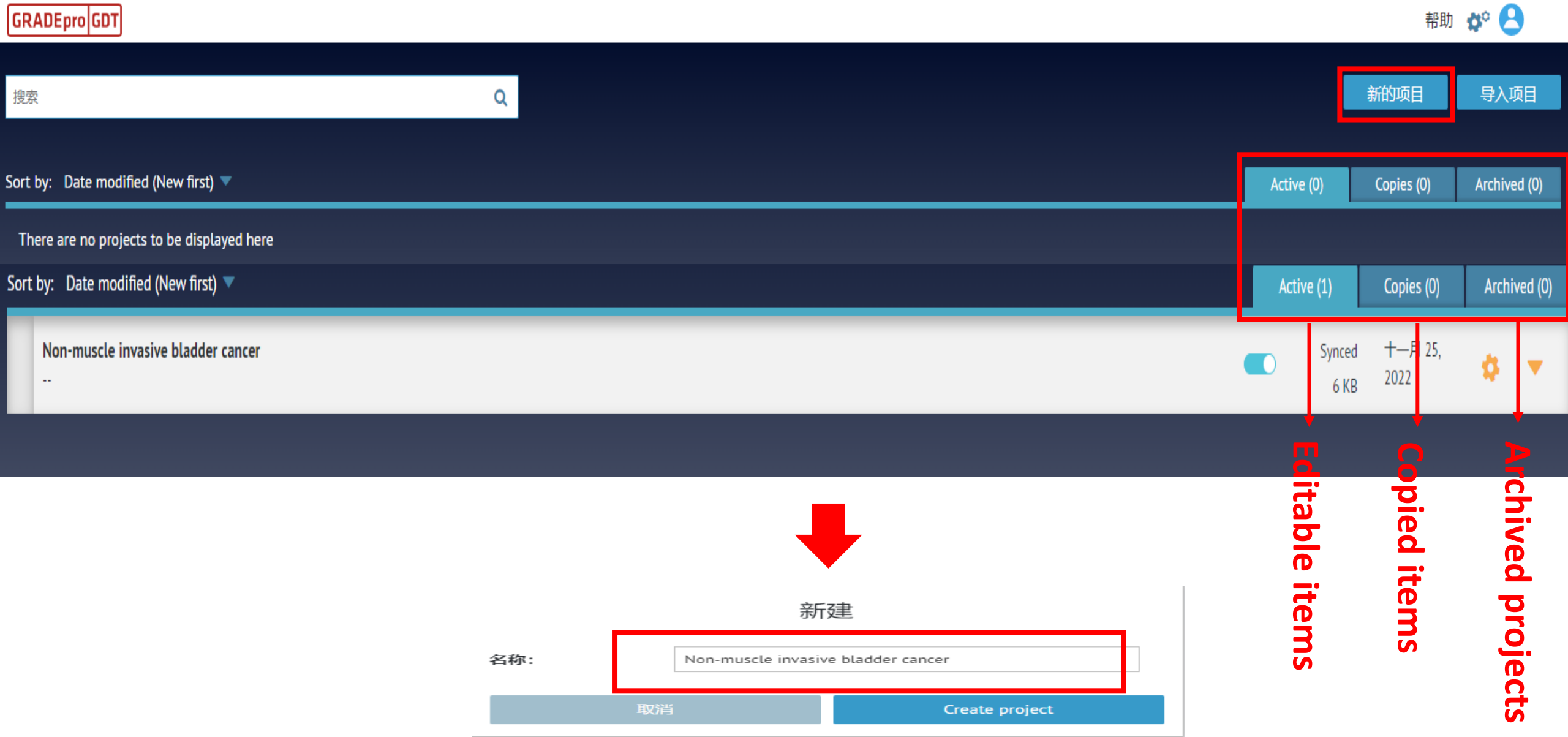
Do not rate up for residual confounding or bias acting against the observed effect.

EXAMPLE

- The initial certainty rating was:
  High certainty: ⨁⨁⨁⨁
- The final rating was:
  Low-certainty evidence: ⨁⨁◯◯

The downgrading decisions were as follows

GRADE domain	Judgment	Adjustment
Risk of bias	Serious	−1
Inconsistency	Serious	−1
Indirectness	Not serious	0
Imprecision	Not serious	0
Publication bias	No important concerns identified	0
Large magnitude of effect	Criterion not met; RR was only 1.25	0
Dose–response gradient	No relevant evidence was reported	0
Residual confounding or bias	No evidence that all plausible residual confounding or bias would reduce the observed effect	0


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帮助
搜索
新的项目
导入项目
Sort by: Date modified (New first)
Active (0)
Copies (0)
Archived (0)
There are no projects to be displayed here
Sort by: Date modified (New first)
Active (1)
Copies (0)
Archived (0)
Non-muscle invasive bladder cancer
--
Synced
6 KB
十一月 25,
2022
Editable items
Copied items
Archived projects

新建
名称:
Non-muscle invasive bladder cancer
取消
Create project

Project setup
任务
团队
范围
References
预后
对照
Multi comparisons
PanelVoice
文件区
传播

Question list 0/12

新组

添加管理【治疗】问题 | 添加诊断问题 | 导入问题

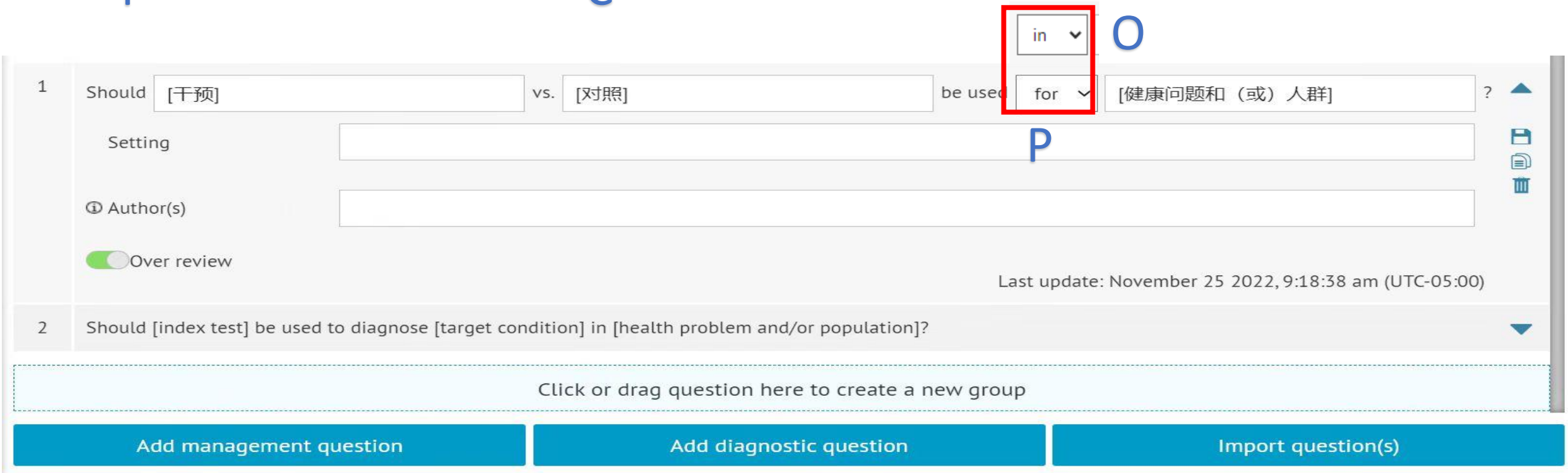
1 与 是否应该用于？
Setting
Author(s)
Over review
Last update: November 25 2022, 9:18:38 am (UTC-05:00)

2 Should [index test] be used to diagnose [target condition] in [health problem and/or population]?

Click or drag question here to create a new group

Add management question | Add diagnostic question | Import question(s)

I C O P


in
1
Should [干预] vs. [对照] be used for [健康问题和（或）人群] ?
Setting
Author(s)
Over review
Last update: November 25 2022, 9:18:38 am (UTC-05:00)
2
Should [index test] be used to diagnose [target condition] in [health problem and/or population]?
Click or drag question here to create a new group
Add management question
Add diagnostic question
Import question(s)

Should 卡介苗单纯诱导 vs. 单纯诱导+维持治疗 be used for 非肌层浸润性膀胱癌患者?

Bottom panel | Explanations

卡介苗单纯诱导 compared to 单纯诱导+维持治疗 for 非肌层浸润性膀胱癌患者

Certainty assessment							Summary of findings			
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Impact	Certainty	Importance	
New outcome										
								-		

Add outcome | Import outcome(s)

New outcome

Short name

Assessed/measured with

Length of follow-up

0

Type

- dichotomous
- continuous
- time to event
- narrative

- pooled
- not pooled
- range of effects
- single study
- not measured
- not reported

Certainty assessment							Summary of findings			
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Impact	Certainty	Importance	

复发率 (follow-up: range 1 weeks to 3 weeks)

18	randomised trials		

№ of studies

18

Study design

randomised trial

Cancel

Risk of bias

not serious
serious
very serious

Assess Risk of Bias

clear

Inconsistency

not serious
serious
very serious

clear

Indirectness

Assess Directness
not serious
serious
very serious

clear

GRADE assessment of study limitations Most information i

No serious limitations, do not downgrade.

Final assessment

Cancel | Apply

Outcome: 复发率

Domain (original question asked	Description	Judgment - Is the evidence sufficiently direct?
Population:		Yes / Probably yes / Probably no / No
Intervention: 卡介苗单纯诱导		Yes / Probably yes / Probably no / No
Comparator: 单纯诱导+维持治疗		Yes / Probably yes / Probably no / No
Direct comparison		Yes / Probably yes / Probably no / No
Outcome: 复发率		Yes / Probably yes / Probably no / No
Final judgment about indirectness across domains:		No indirectness / Serious indirectness / Very serious indirectness

Cancel | Apply

Certainty assessment							Summary of findings			
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Impact	Certainty	Importance	
复发率 (follow-up: range 1 weeks to 3 weeks)										
18	randomised trials	not serious	serious	not serious	serious	none		⨁⨁◯◯ Low		

Add outcome | Import outcome(s)

Other considerations

Publication bias: [ ]
Large effect: [no]
Plausible confounding: [no]
Dose response gradient: [no]

Cancel | Apply

[ ]
- undetected
- strongly suspected

clear

[no]
- no
- large
- very large

[no]
- no
- would reduce demonstrated effect
- would suggest spurious effect

clear

[no]
- no
- yes

clear

Narrative description

Cancel | Apply

Importance

- 9 - critical
- 8 - critical
- 7 - critical
- 6 - important
- 5 - important
- 4 - important
- 3 - not important
- 2 - not important
- 1 - not important

Certainty assessment							Summary of findings					
							Impact				Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	№ of patients		Effect			
							[intervention]	[comparison]	Relative (95% CI)	Absolute (95% CI)		
复发率 (follow-up: range 1 weeks to 3 weeks)												
18	randomised trials	not serious	serious	not serious	serious	none	60/100 (60.0%)	24/100 (24.0%)	**RR 2.5** (1.8 to 2.9)	**360 more per 1,000** (from 192 more to 456 more)	⨁⨁◯◯ Low	CRITICAL

Add outcome

Intervention

With event

Total

Cancel | Apply

Control

With event: 24

Total: 100

Calculated control: 24.0 %

Control risk

Low: Low

Moderate: Moderate

High: High 30 %

Cancel | Apply

Relative effect

Relative | of

95% CI from | to

Cancel | Apply

Absolute effect

Absolute effect auto calculation

Absolute: 192 more per 1,000

95% CI from 48 more to 312 more

Cancel | Apply

EXAMPLE

Export GRADE evidence profile

Choose up to 7 outcomes

Select all

- [x] 复发率 (CRITICAL)
- [x] New outcome

Choose file format

- .sof (for import to RevMan)
- MS Word
- HTML (for non MS Office users)
- PDF
- JSON-LD

Choose orientation of the table:

- Landscape
- Portrait

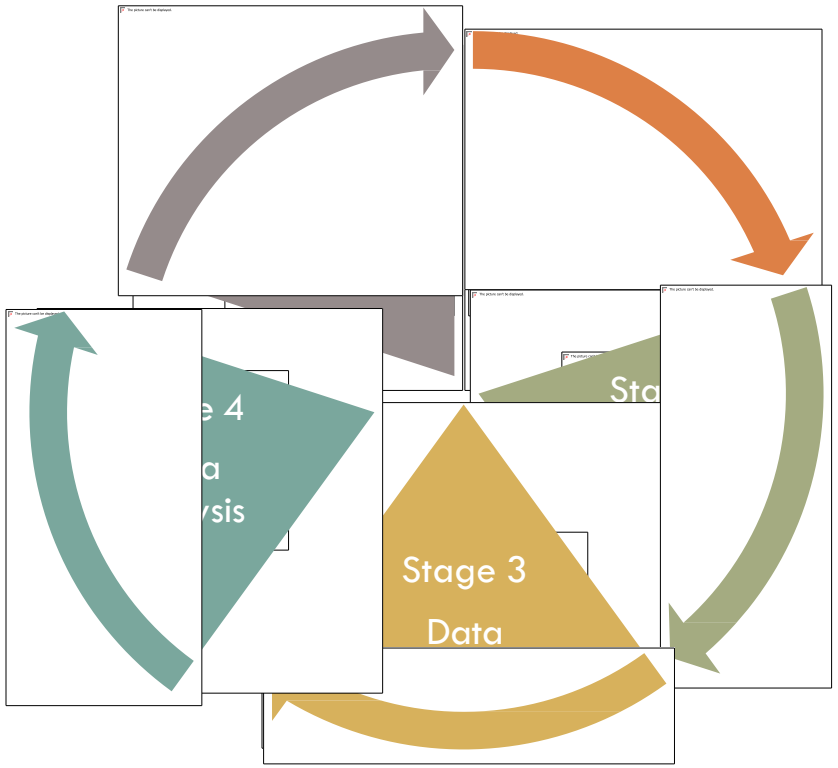
Cancel | Download

Author(s): Xiaohua Zhang
Question: [redacted] compared to [redacted]+[redacted] for [redacted]
Setting: ICU
Bibliography:

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	[redacted]	[redacted]+[redacted]	Relative (95% CI)	Absolute (95% CI)		
[redacted] (follow-up: range 1 weeks to 3 weeks)												
18	randomised trials	not serious	serious	not serious	serious	none	60/100 (60.0%)	24/100 (24.0%)	**RR 2.5** (1.8 to 2.9)	**360 more per 1,000** (from 192 more to 456 more)	⨁⨁◯◯ Low	CRITICAL

CI: confidence interval; **RR:** risk ratio

Stage 4: Data Analysis



Important concepts in Meta-analysis



Effect size/ Effect estimate



- **Summary statistic**
- **Statistical concept** that measures the **strength of the relationship** between two variables on a numeric scale
- **Quantitative measure** of the magnitude of the effect

Effect size/Summary statistic for each study

For binary data

- Ratio of risks - Risk ratio, Relative risk
- Difference in risks - Risk difference
- Ratio of odds - Odds ratio

For continuous data

- Standardised measures- Standardized mean difference (SMD)
- Unstandardised measures- Mean difference

Time to event

- Hazard ratio

Standardised Mean difference (SMD)

Cohens d

$$Cohen's\ d = \frac{x_1 - x_2}{s_{pooled}}$$

$$s_{pooled} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Hedges g

$$d = c(m) X Cohen's\ d)$$

c(m)=1-3/(4m-1) with

$$m = n_1 + n_2 - 2$$

RULE OF THUMB

- Small effect = 0.2
- Medium Effect = 0.5
- Large Effect = 0.8

Weight



- In a meta-analysis, more weight to the studies which give us more information
- Weight assignment depends on **sample size**
- Weight is proportional to **inverse of variance**
- Studies with **narrow** 95% CI is given more weight

$$\text{Weight} = \frac{1}{\text{Variance}}$$

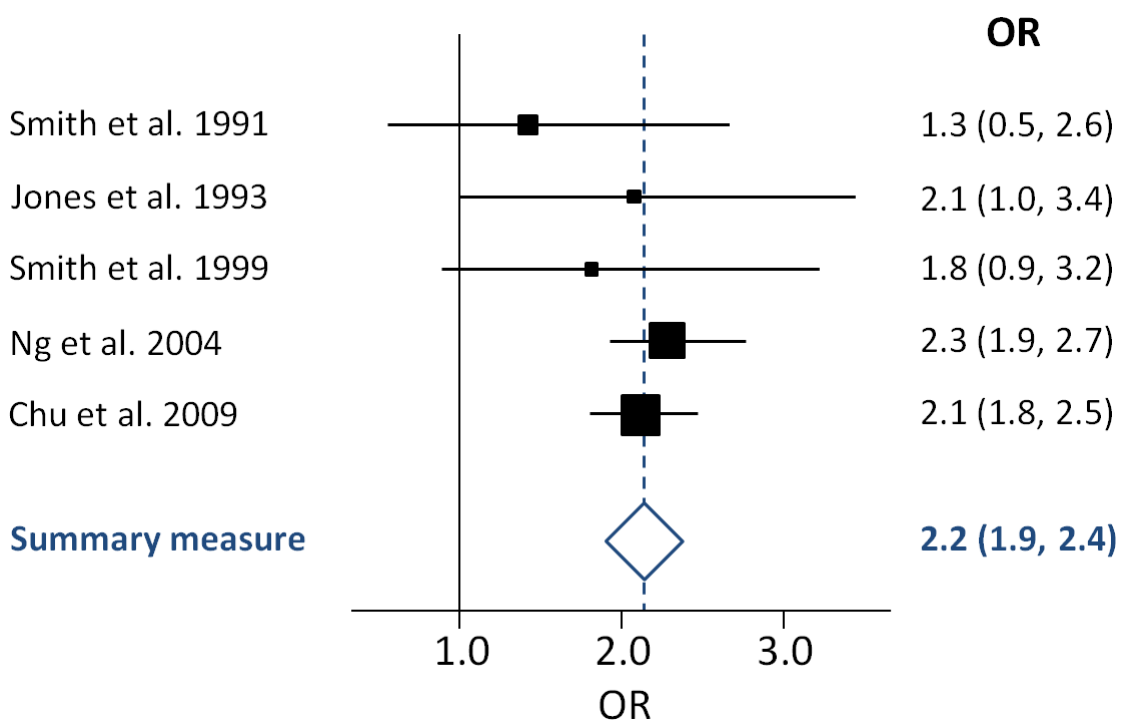
Pooled effect estimate
(Inverse variance method)

$$\overline{X} = \frac{\sum W_i X_i}{\sum W_i}$$

	Effect Size (Xi)	Variance (Vi)	Weight (1/Vi)	W x Xi
Study 1	x1	v1		
Study 2	x2	v2		
Study 3	x3	v3		
			∑wi =	∑wi.xi =
				$\overline{X} = \frac{\sum W_i X_i}{\sum W_i}$

Forest plot

- Forest of lines
- **Blobbogram**

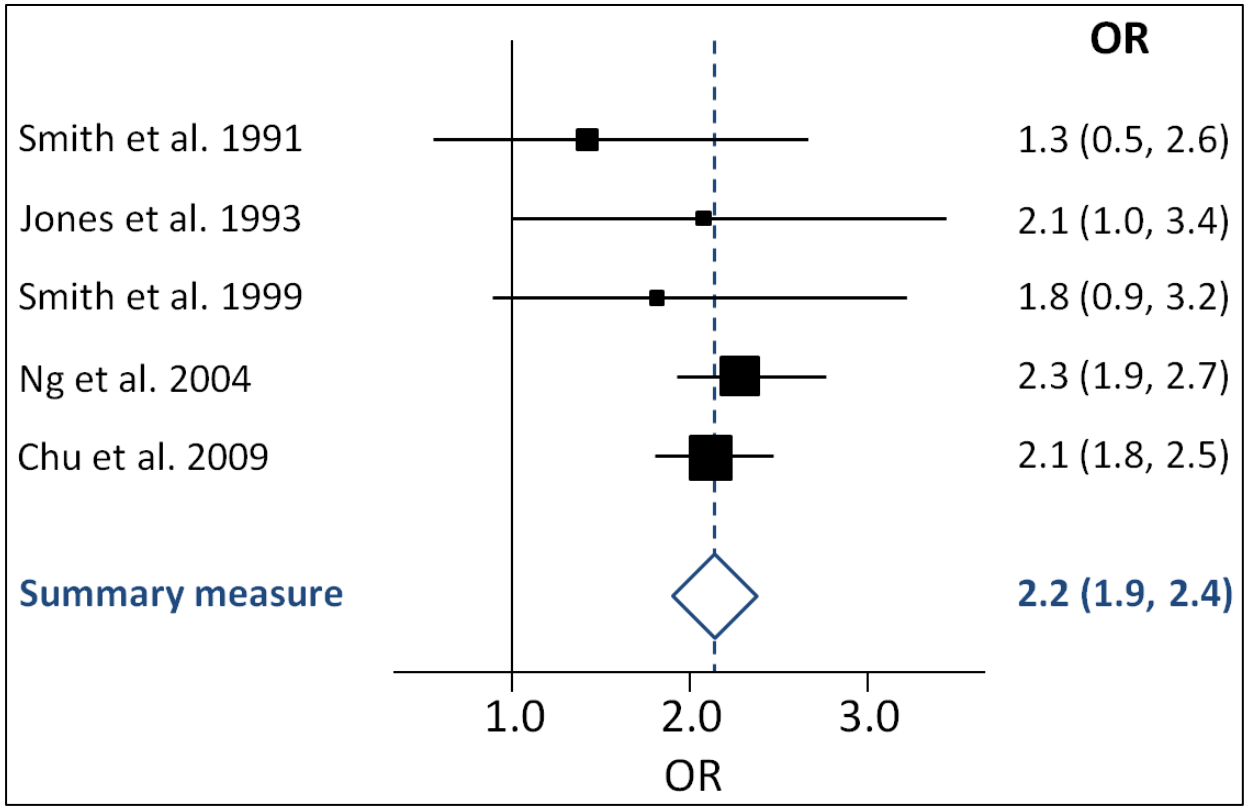


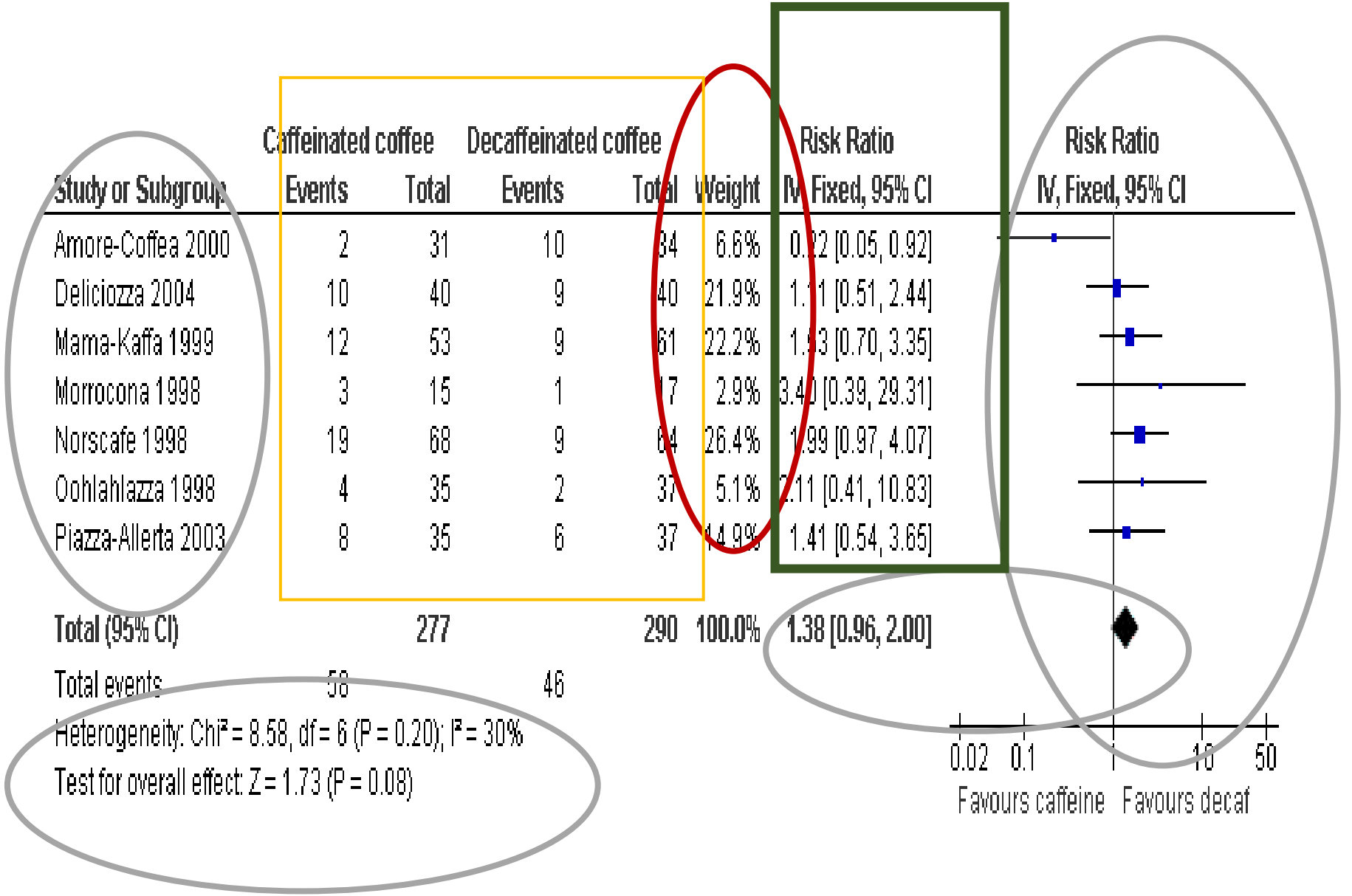


- Effect estimates from each study
- Combined or pooled estimate

Forest plot

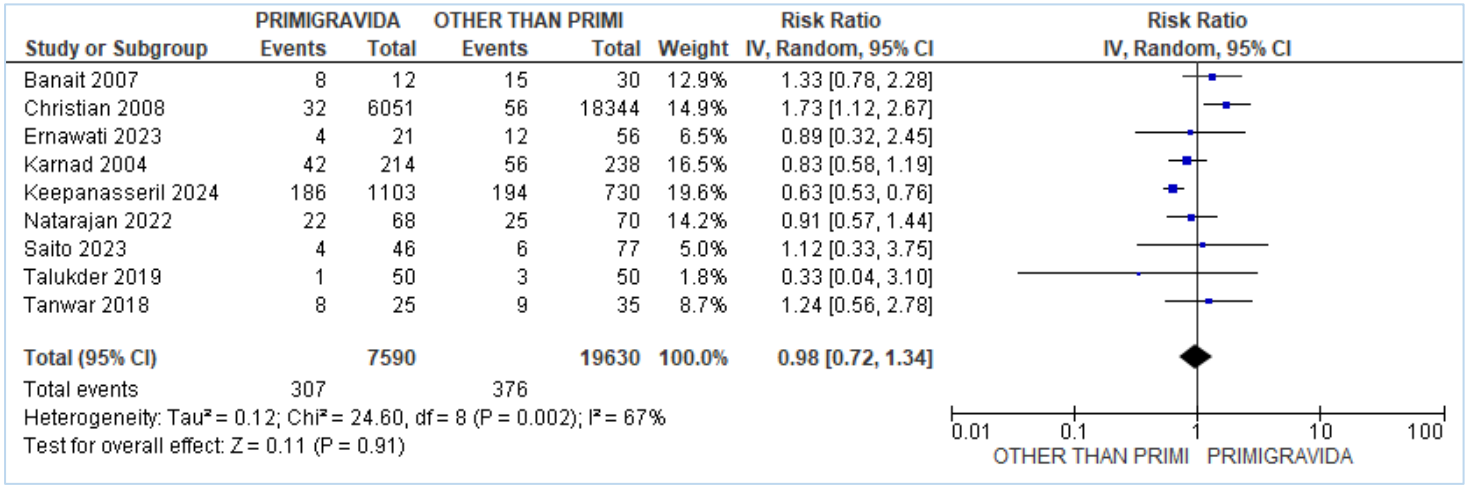
- Each study represented by a box and a horizontal line
  - Midpoint- point estimate
  - Width of line- CI
  - Area of square- weight of each study
- Diamond – overall effect estimate
- Width of diamond - CI



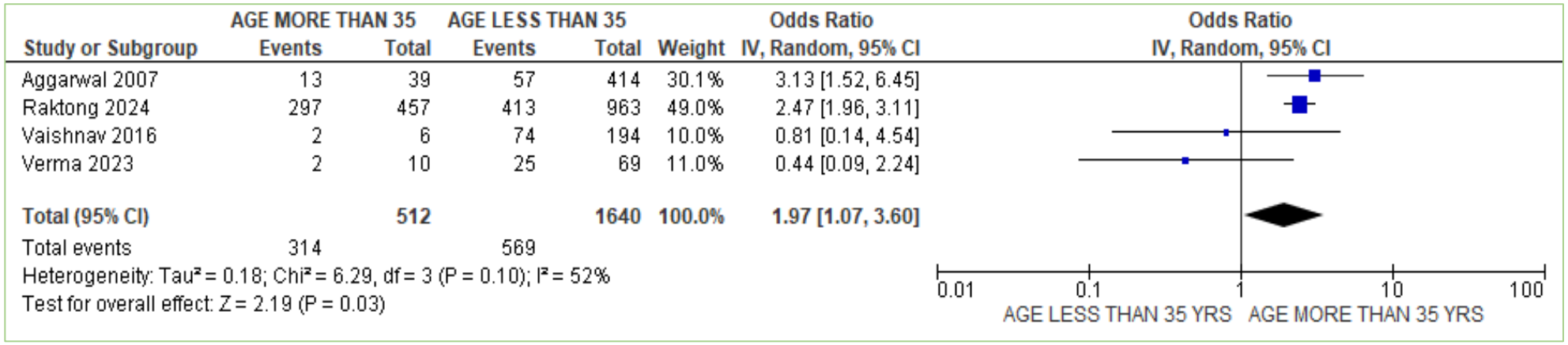

Caffeinated coffee
Decaffeinated coffee
Risk Ratio
Risk Ratio
Study or Subgroup
Events
Total
Events
Total
Weight
M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
Amore-Coffea 2000 2 31 10 34 6.6% 0.22 [0.05, 0.92]
Deliciozza 2004 10 40 9 40 21.9% 1.11 [0.51, 2.44]
Mama-Kaffa 1999 12 53 9 61 22.2% 1.53 [0.70, 3.35]
Morrocona 1998 3 15 1 17 2.9% 3.40 [0.39, 29.31]
Norscafe 1998 19 68 9 64 26.4% 1.99 [0.97, 4.07]
Oohlahlazza 1998 4 35 2 37 5.1% 2.11 [0.41, 10.83]
Piazza-Allerta 2003 8 35 6 37 14.9% 1.41 [0.54, 3.65]
Total (95% CI) 277 290 100.0% 1.38 [0.96, 2.00]
Total events 58 46
Heterogeneity: Chi² = 8.58, df = 6 (P = 0.20); I² = 30%
Test for overall effect: Z = 1.73 (P = 0.08)
0.02
0.1
1
10
50
Favours caffeine
Favours decaf

Association of anaemia complicating pregnancy, maternal age and parity with maternal mortality in the WHO South-East Asian region: a Systematic Review and Meta-analysis of observational studies



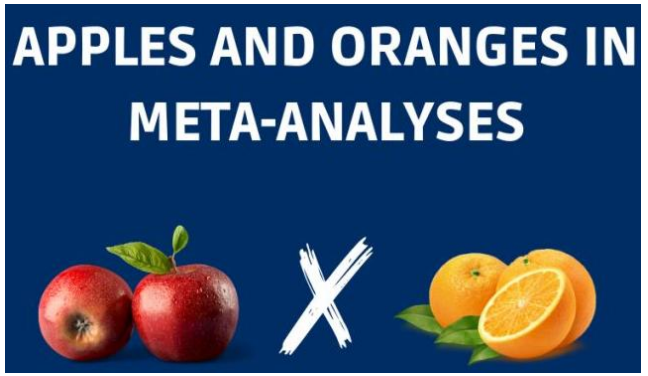


Forest-plot showing the association of primigravida and maternal mortality for cohort studies



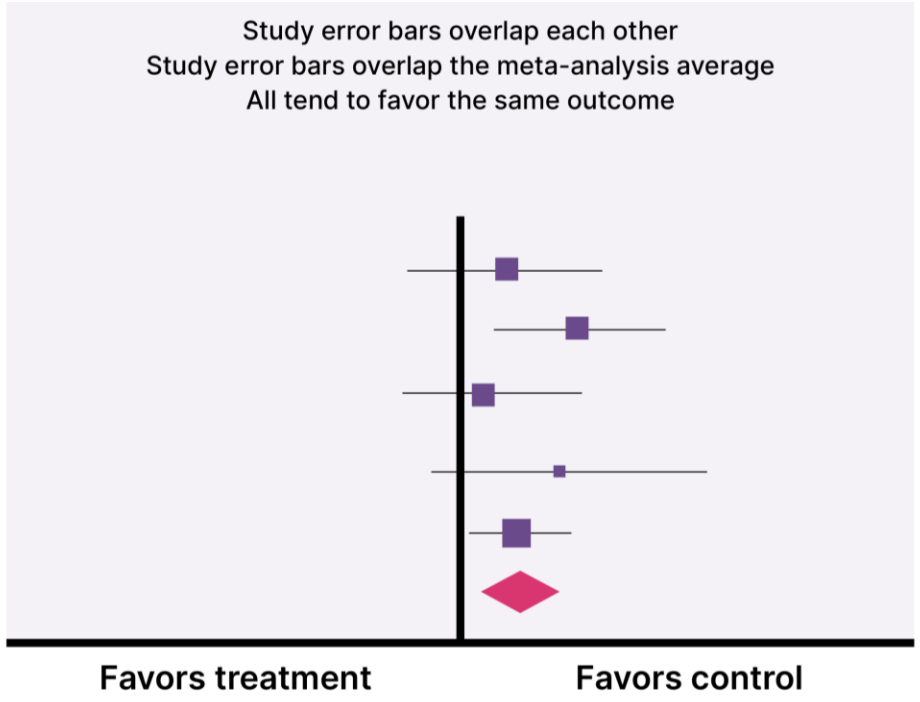
Forest plot showing the association of age more than 35 years and maternal mortality for case-control studies

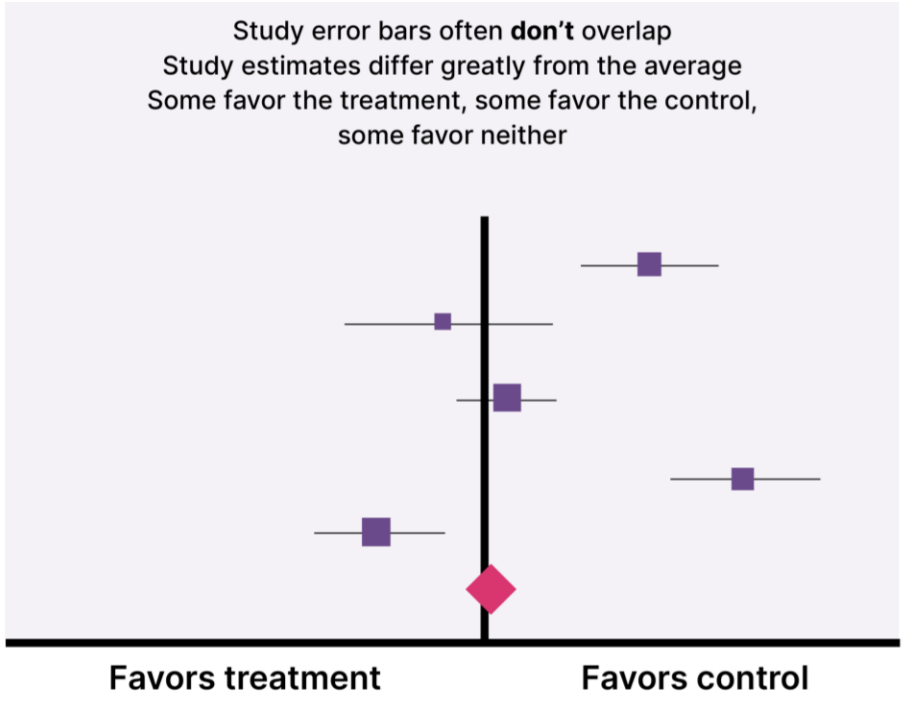
What is Heterogeneity?



- Heterogeneity refers **to any type of variations** that exists **between the studies** included in the SR/MA
- It may be **inevitable** in these types of analysis, since SR/MA brings together multiple studies conducted in different settings
- Visually, it can be identified by looking at the **point estimates** varying widely between the studies and **their confidence interval** not overlapping

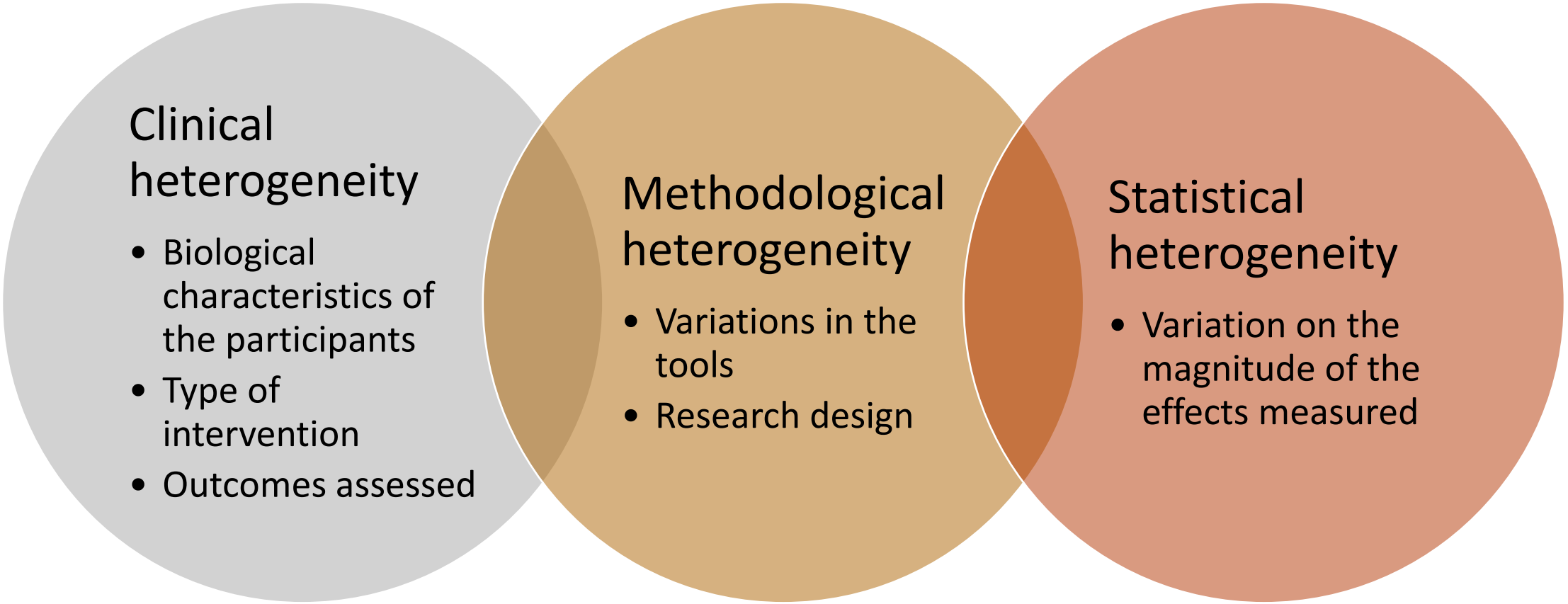
Heterogeneity





High heterogeneity

Heterogeneity- types



Tests for heterogeneity



- Cochrane Q
- I^2
- τ^2

Test for heterogeneity	Value
Q	362.77
p(Q)	0.001
I^2	0.97
τ^2	1.31

Q-statistic ("Cochrane's Q")

$$Q = \sum_{k=1}^{K} w_k(\theta_k - \theta)^2$$

- Based on Chi-square statistic
- Weighted sum of squared differences between the observed effects and the weighted average effect
- Null hypothesis- All the studies are homogeneous
- **p value <0.05 (low p value) - Shows heterogeneity of studies**
- **p value >0.05 (high p-value) - Shows homogeneity of studies**
- Low power (when fewer studies)
- Final interpretation based on other measures of heterogeneity also

I^2 test

- More informative
- A relative measure
- Expressed as a **percentage with a range from 0 to 100 percent**
- The I^2 statistic tells **what percentage of the observed variability in effect estimates is due to heterogeneity across studies (above chance)**

I^2 test

$$I^2 = 100\% \times (Q - df)/Q$$

Q: chi-squared statistic
df: degree of freedom of Q statistic

"Rule of thumb" to interpret I^2 (Higgins et al. 2003):

- I^2 = 25%: low heterogeneity
- I^2 = 50%: moderate heterogeneity
- I^2 = 75%: substantial heterogeneity

0-40%: might not be important

30-60%: moderate heterogeneity

50-90%: substantial heterogeneity

75-100% considerable heterogeneity

ICMR handbook

I^2 test

- **I^2** -used as a criterion for a decision whether a **subgroup analysis** is indicated
  - If **I^2 is low, then there is no heterogeneity** (no need of subgroup analysis)
  - If **I^2 is large, suggest heterogeneity** (need of subgroup analysis)

τ^2 (tau squared)

- Absolute value- Reflects the **between-study variance**
- τ^2 is not used itself as a measure of heterogeneity - value does not have much interpretation
- τ^2 **closer to zero - low variability** between studies
- Larger values indicate greater true heterogeneity
- **Calculated in REM**

Tests for heterogeneity

Test for heterogeneity	Value
Q	362.77
p(Q)	0.001
I^2	0.97
τ^2	1.31

Graphical methods for identification of heterogeneity

L’ Abbé plot

Galbraith plot

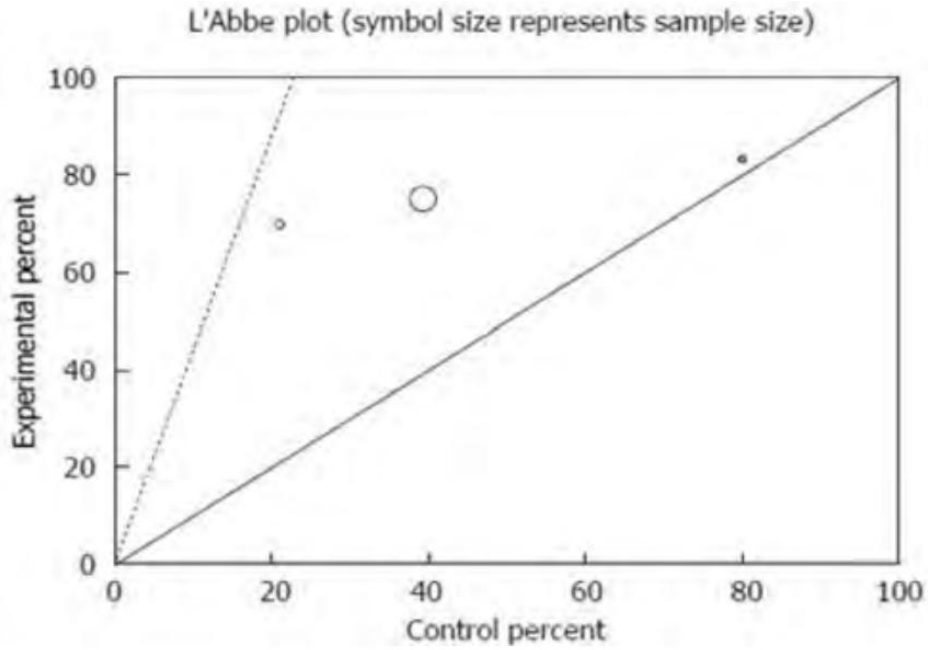
Baujat plot

Funnel plot

Forest plot

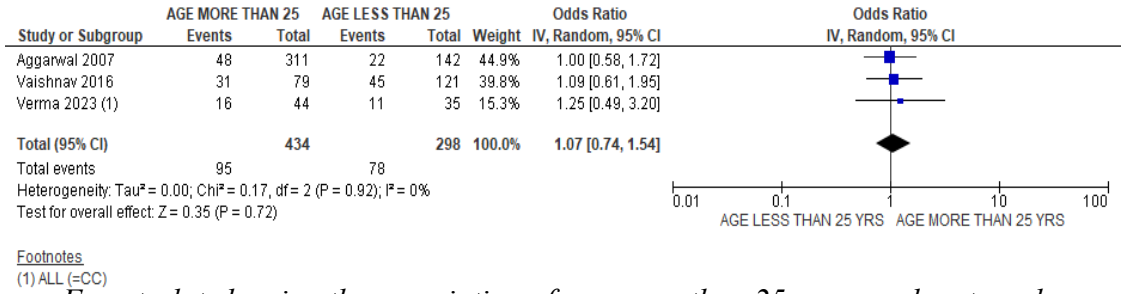
i. L’ Abbé plot

- X axis- observed event rate in the control group
- Y axis- observed event rate in the experimental group
- The size of each point is typically proportional to the study's weight or sample size.
- **A large spread of points** far from the overall effect line may signal significant heterogeneity among the studies.



i. L’ Abbé plot

Association of anaemia complicating pregnancy, maternal age and parity with maternal mortality in the WHO South-East Asian region: a Systematic Review and Meta-analysis of observational studies



Study or Subgroup	AGE MORE THAN 25 Events	Total	AGE LESS THAN 25 Events	Total	Weight	Odds Ratio IV, Random, 95% CI
Aggarwal 2007	48	311	22	142	44.9%	1.00 [0.58, 1.72]
Vaishnav 2016	31	79	45	121	39.8%	1.09 [0.61, 1.95]
Verma 2023 (1)	16	44	11	35	15.3%	1.25 [0.49, 3.20]
Total (95% CI)		**434**		**298**	**100.0%**	**1.07 [0.74, 1.54]**
Total events	95		78			

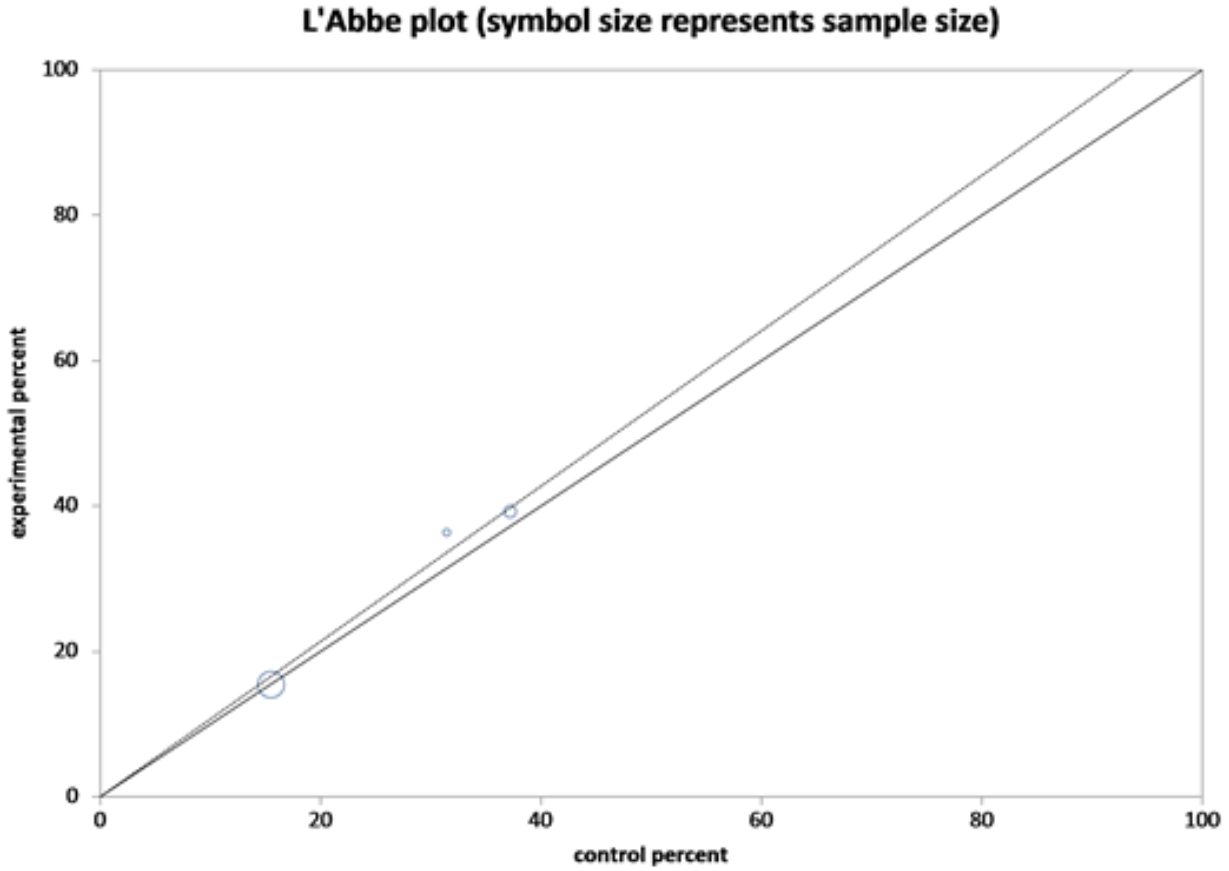
Heterogeneity: Tau² = 0.00; Chi² = 0.17, df = 2 (P = 0.92); I² = 0%
Test for overall effect: Z = 0.35 (P = 0.72)

Odds Ratio IV, Random, 95% CI

0.01 0.1 1 10 100
AGE LESS THAN 25 YRS AGE MORE THAN 25 YRS

Footnotes
(1) ALL (=CC)

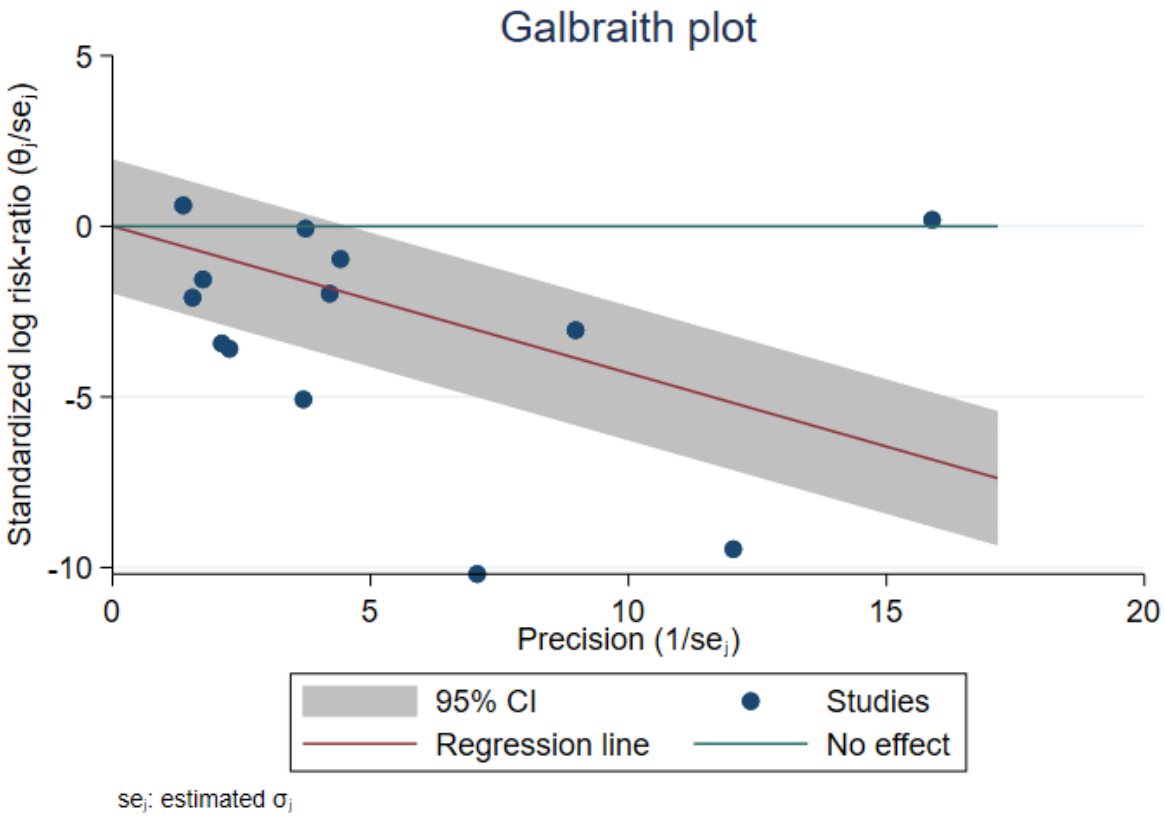
Forest-plot showing the association of age more than 25 years and maternal mortality (all case-control studies)



L’Abbe plot showing the heterogeneity between studies with ‘age>/= 25 years” data

ii. Galbraith plot

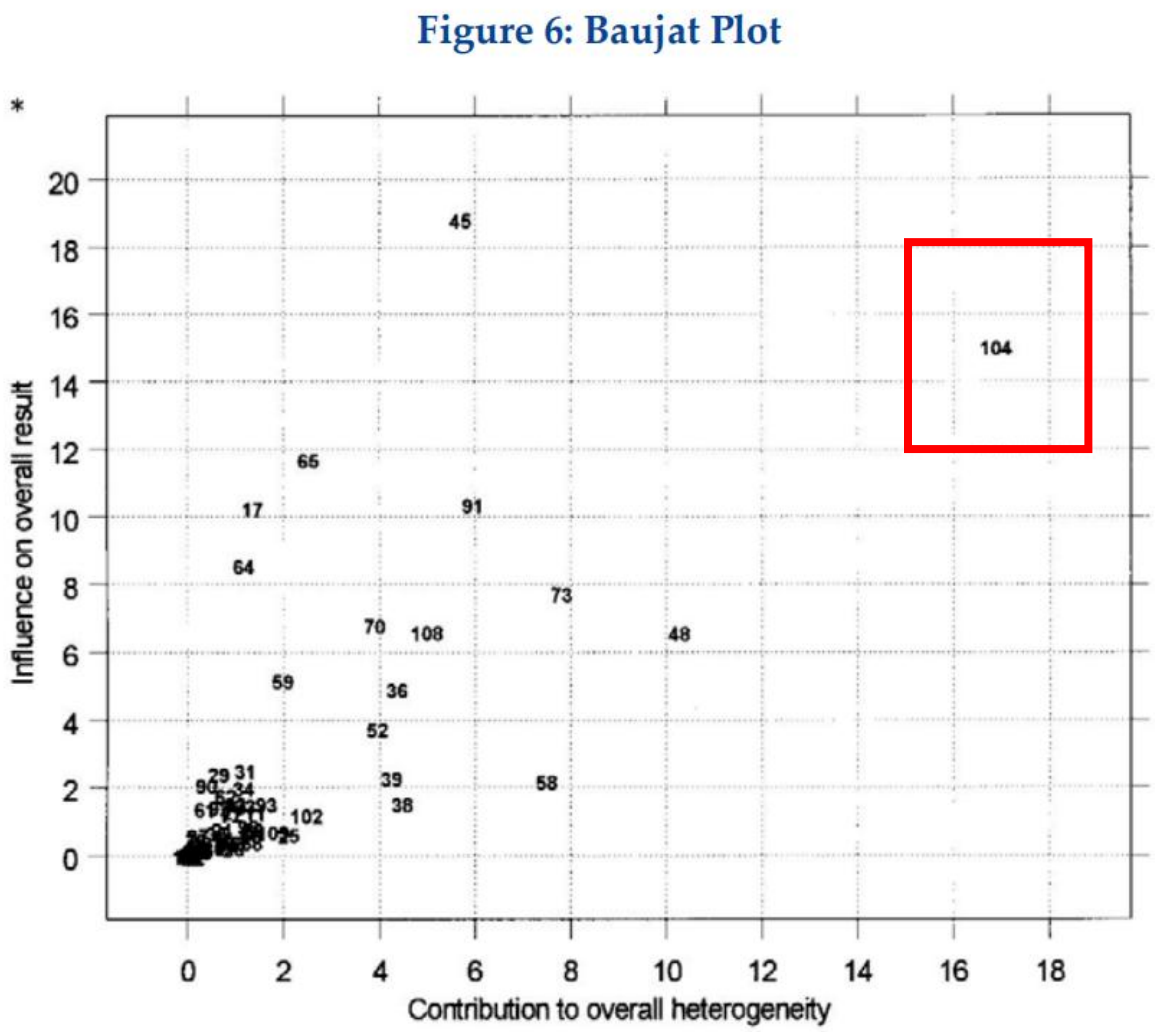
- X axis- precision (inverse Std Error)
- Y axis – standardized effect size or z score (effect size divided by its Std Error)
- Galbraith plots facilitate examination of **heterogeneity**, including detection of **outliers**



- The degree of dispersion of all data points around the red regression line.

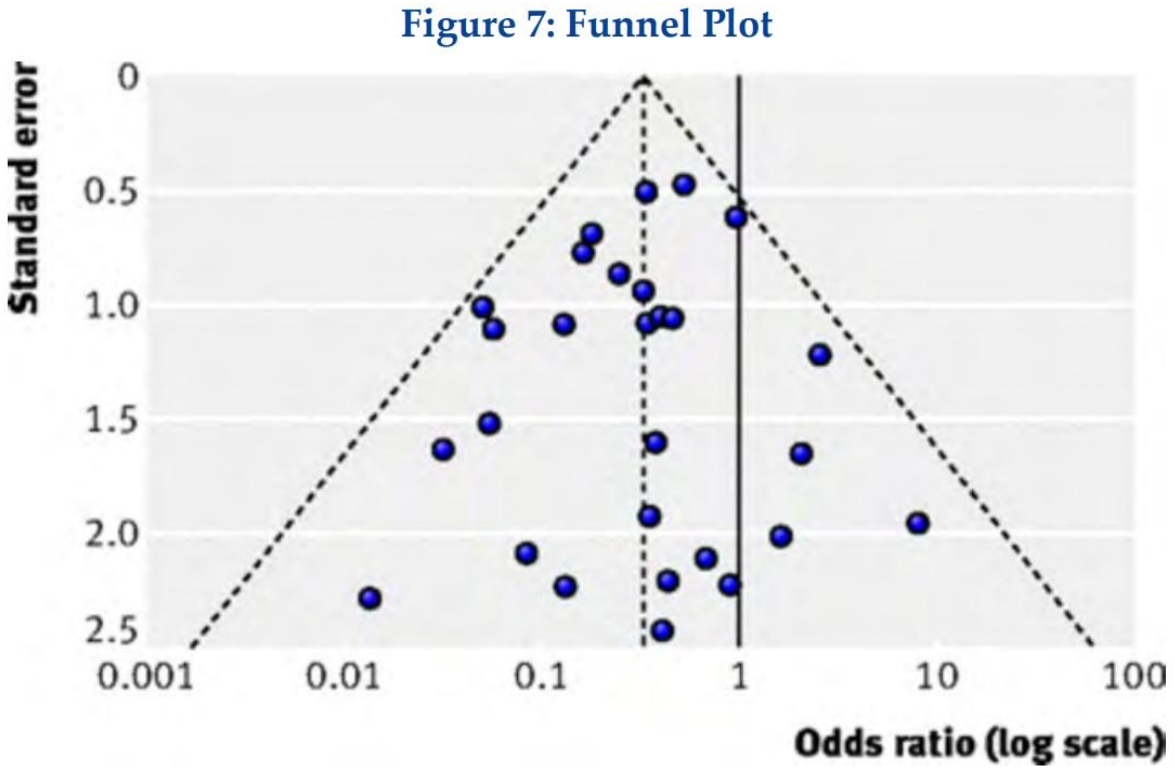
iii. Baujat plot

- X axis - the contribution of the study to the overall Cochran Q-test for heterogeneity
- Y axis - the influence of the study on overall effect
- Studies in the **top-right corner** are **high-impact outliers**, potentially skewing the results- exclude those



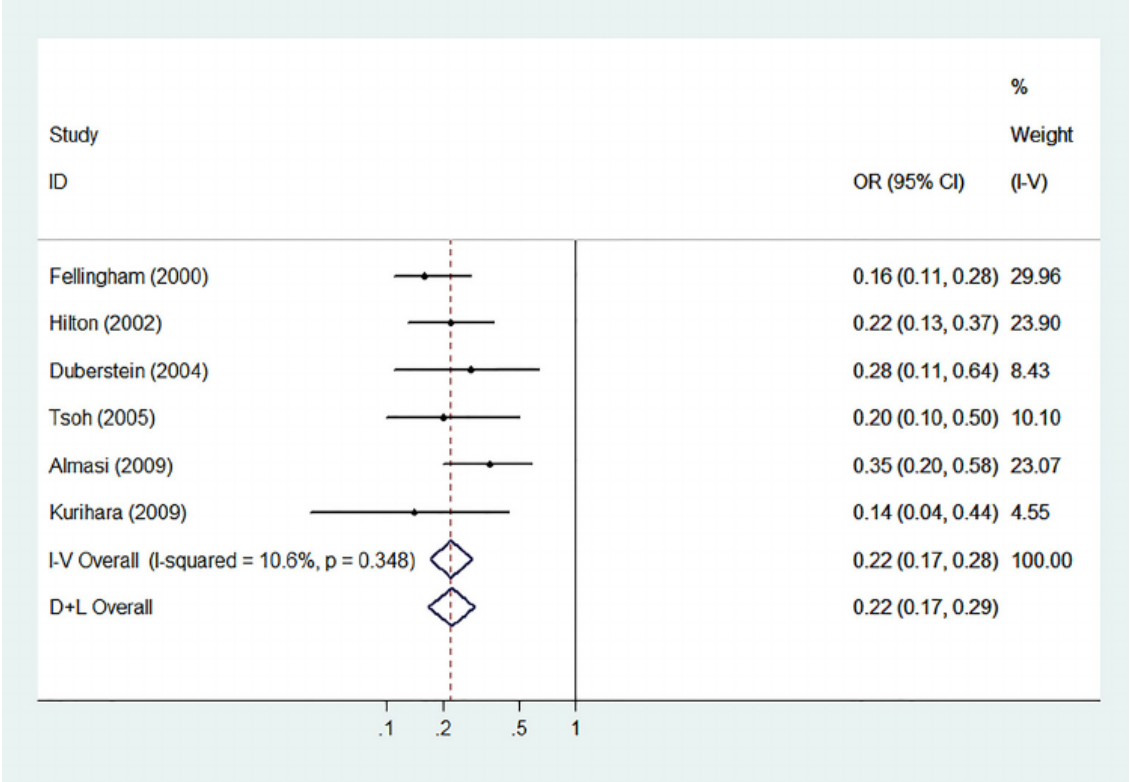
iv. Funnel plot

- X axis- effect estimate
- Y axis- measure of precision (or study size)
- For heterogeneity
- Primarily used as a visual aid for detecting **publication bias**



v. Forest plot

- Greater overlap between the confidence intervals indicates less heterogeneity



Models for Statistical analysis



‘Fixed effect’ model

‘Random effects’ model

- **Fixed effect model** - it is assumed that there is “no heterogeneity”
- **Random effects model** - it is assumed that there is “heterogeneity”

Models for Statistical analysis

Fixed Effects Model (FEM)

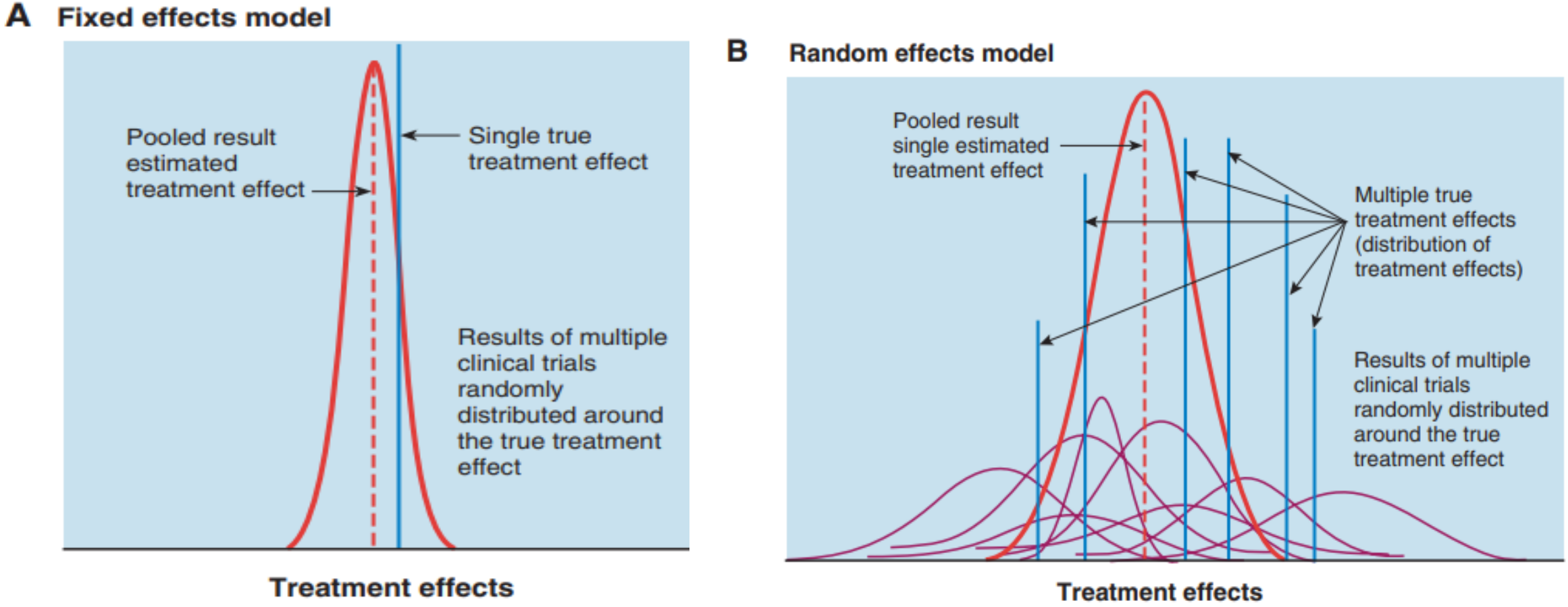
- “Fixed effect” - only one underlying effect size for all the studies - **only one true effect**
- The basic assumptions are:
  - Studies use **identical methods, patients, and measurements;**
  - Produce **identical results**;
  - Differences are only due to **within-study variation.**
- All of the observed differences between studies is due to chance or due to sampling error
- Can be used- When the studies are considered to have the **same design and methodology**

Models for Statistical analysis

Random Effects Model (REM)

- Assumption: The random effects model assumes **study populations vary** and thus true effects differ from each other
- i.e, The true effect could vary from study to study
- Variability of the estimate may have two sources: **within the study and between the studies**

Models for Statistical analysis



Characteristics of Fixed VS Random Effects models

	Fixed Effects	Random Effects
Pooled estimate weighted by	Within study variance	Within and between study variance
Weighting of small studies	Smaller weights	Larger weights
Weighting of large studies	Larger weights	Smaller weights
Confidence intervals (generally)	Narrow (within study variance only)	Wide (due to including variance between sudies)
In the presence of significant/substantial statistical heterogeneity	Narrow (do not reasonably account for the variance in effect estimates between the studies)	Even wider
Effect size (pooled summary effect)	Estimate of some common effect size; use Z	Estimate of the mean of a normal distribution of effect sizes: use t distrib

Choice of models

- If you are looking for a single best effect estimate, and the studies appear to be homogeneous, then a fixed effects model is preferable

- If there is evidence of heterogeneity that cannot be explained
  - Random effects approach is preferrable
  - But … consider

- If studies are not likely functionally equivalent, and goal is to generalize to a range of scenerios
  - Random effects

Methods for combining study estimates

Fixed Effects Methods

- Methods:
  - Small number of studies, but sample sizes of the studies are large
    - Inverse variance
      - Preferred
  - If sparse data
    - Mantel & Haenszel method (RR, OR etc)
      - Preferred if many studies , each being small
      - Use continuity correction (add 0.5 to a cell) if cell is 0
    - Peto method for OR
      - Can be used if you have 0 in a cell of the 2x2 table
      - Produces serious underestimates when OR is large
      - Many criticisms
  - For large sample sizes
    - Maximum likelihood method

Methods for combining study estimates

Random Effects Methods

- Weighted least squares (WLS) regression
  - Inverse variance weighting
  - Most common method
  - Preferred
  - Dersimonian & Laird method
    - Method of moments estimate for tau-sq
- Maximum likelihood method (ML)
  - Assumes normality of the underlying effect distribution
    - Has another estimate for tau-sq
- Restricted maximum likelihood method (REML)
  - Assumes normality of the underlying effect distribution
- Exact method suggested by Van Houwelingen
  - Non-parametric approach if normality assumption is violated

The Inverse Variance Weight

- Studies generally vary in size.
- An ES based on 100 subjects is assumed to be a more "precise" estimate of the population ES than is an ES based on 10 subjects.
- Therefore, larger studies should carry more "weight" in our analyses than smaller studies.
- Simple approach: weight each ES by its sample size.
- Better approach: weight by the inverse variance.

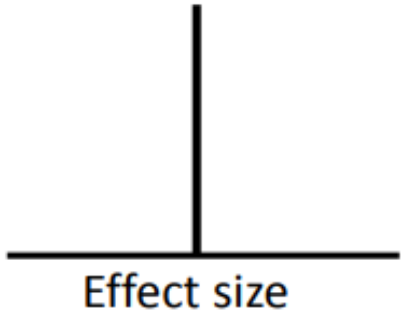
Weighting under models

- Fixed effect model

$$\text{Weight} = \frac{1}{\text{variance within}} = \frac{1}{SE^2}$$

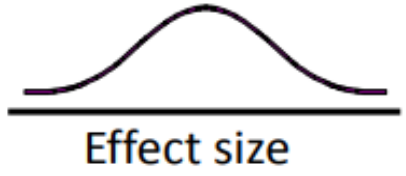
- Random effect model

$$\text{Weight} = \frac{1}{(\text{variance within} + \text{variance between})} = \frac{1}{SE^2 + Tau^2}$$



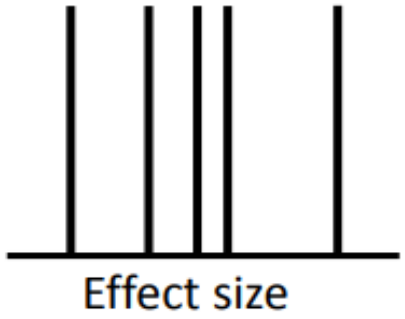
The underlying effect sizes are identical

Common-effect model
(or fixed-effect model)



The underlying effect size are different and related through some distribution

Random-effects model

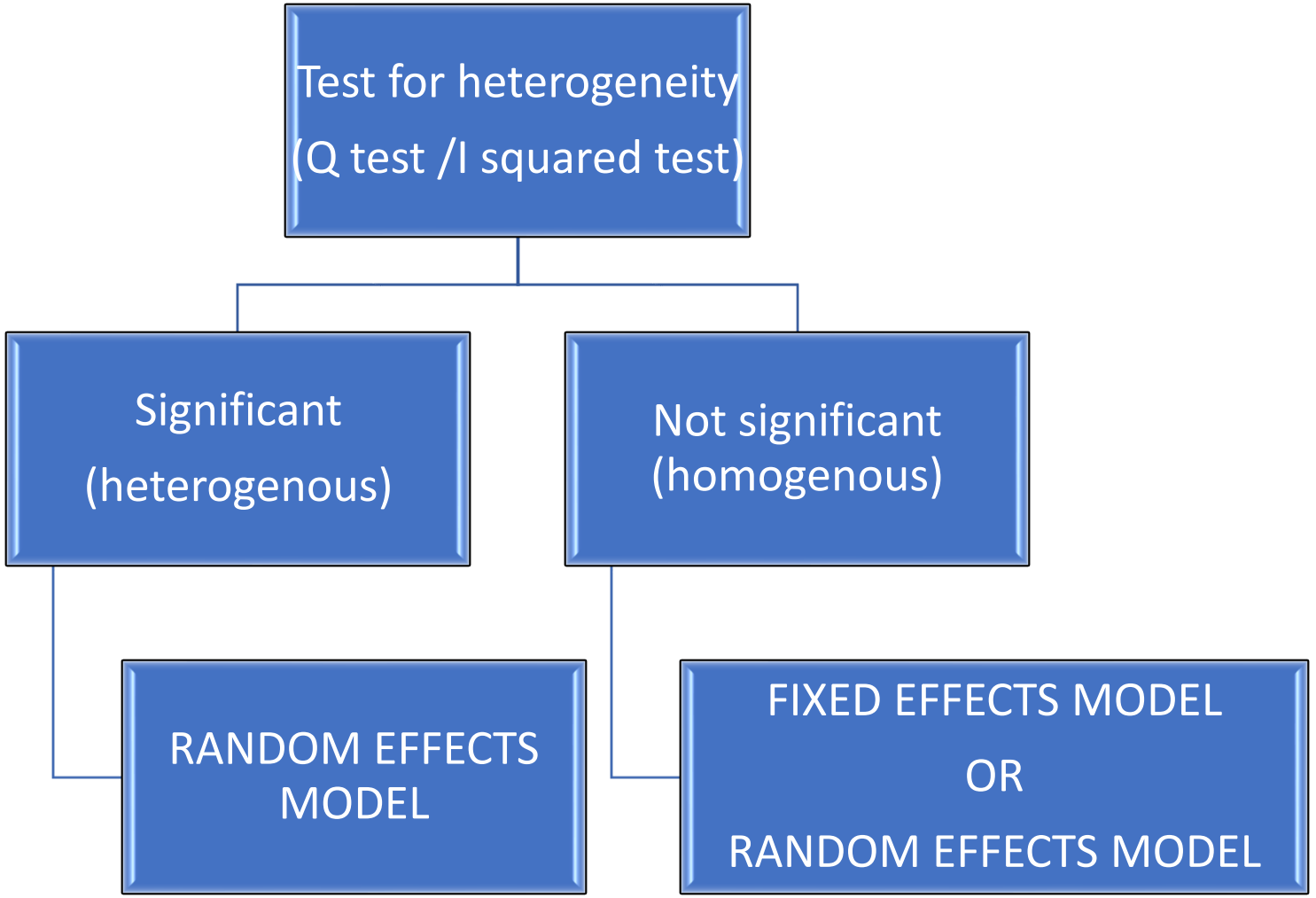


The underlying effect sizes are different and unrelated (independent)

Fixed-effects model

FEM or REM

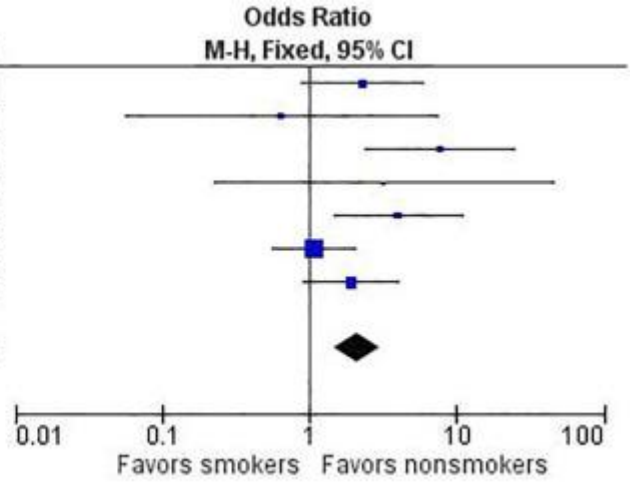
- Based on heterogeneity



FEM or REM?????

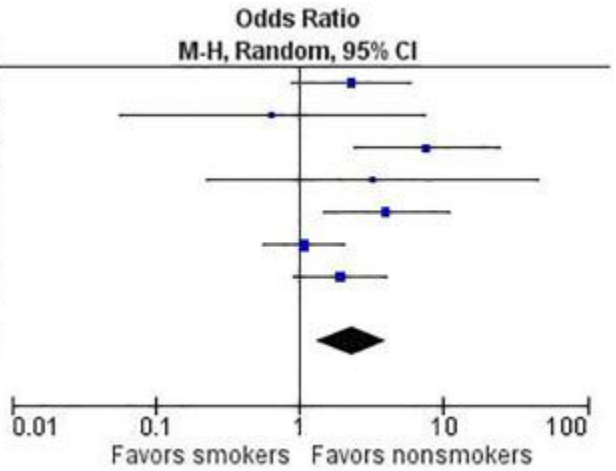
Study or Subgroup	Smokers Events	Smokers Total	Nonsmokers Events	Nonsmokers Total	Weight	Odds Ratio M-H, Fixed, 95% CI
An 1995	16	34	12	43	13.9%	2.30 [0.89, 5.92]
Bose 2001	1	46	2	60	4.2%	0.64 [0.06, 7.33]
Brown 1986	20	50	4	50	5.9%	7.67 [2.38, 24.65]
Emery 1997	2	3	5	13	1.5%	3.20 [0.23, 45.19]
Goldberg 2002	15	30	10	50	9.3%	4.00 [1.48, 10.83]
Luszczyk 2013	14	156	35	417	42.9%	1.08 [0.56, 2.06]
Martin 1999	13	75	21	214	22.3%	1.93 [0.91, 4.07]
Total (95% CI)		**394**		**847**	**100.0%**	**2.11 [1.49, 3.00]**
Total events	81		89			

Heterogeneity: Chi² = 11.50, df = 6 (P = 0.07); I² = 48%
Test for overall effect: Z = 4.19 (P < 0.0001)



Study or Subgroup	Smokers Events	Smokers Total	Nonsmokers Events	Nonsmokers Total	Weight	Odds Ratio M-H, Random, 95% CI
An 1995	16	34	12	43	17.2%	2.30 [0.89, 5.92]
Bose 2001	1	46	2	60	4.6%	0.64 [0.06, 7.33]
Brown 1986	20	50	4	50	13.7%	7.67 [2.38, 24.65]
Emery 1997	2	3	5	13	4.0%	3.20 [0.23, 45.19]
Goldberg 2002	15	30	10	50	16.3%	4.00 [1.48, 10.83]
Luszczyk 2013	14	156	35	417	23.1%	1.08 [0.56, 2.06]
Martin 1999	13	75	21	214	21.0%	1.93 [0.91, 4.07]
Total (95% CI)		**394**		**847**	**100.0%**	**2.29 [1.30, 4.03]**
Total events	81		89			

Heterogeneity: Tau² = 0.25; Chi² = 11.50, df = 6 (P = 0.07); I² = 48%
Test for overall effect: Z = 2.88 (P = 0.004)



Methods for analysis

TYPE OF DATA	EFFECT MEASURE	FIXED EFFECT METHODS	RANDOM EFFECT METHODS
Dichotomous	Odds ratio	Mantel- Haenzel Inverse variance Peto	Mantel- Haenzel Inverse variance Der-Simonian Liard
	Risk ratio	Mantel- Haenzel	Mantel- Haenzel Der-Simonian Liard
	Risk difference	Inverse variance	Inverse variance Der-Simonian Liard
Continuous	Mean difference	Inverse variance	Inverse variance
	Standardised mean difference	Inverse variance	Inverse variance

Analysing reasons for heterogeneity

- If heterogeneity is found, then various methods could be adopted for investigating the possible reasons

Sensitivity analysis

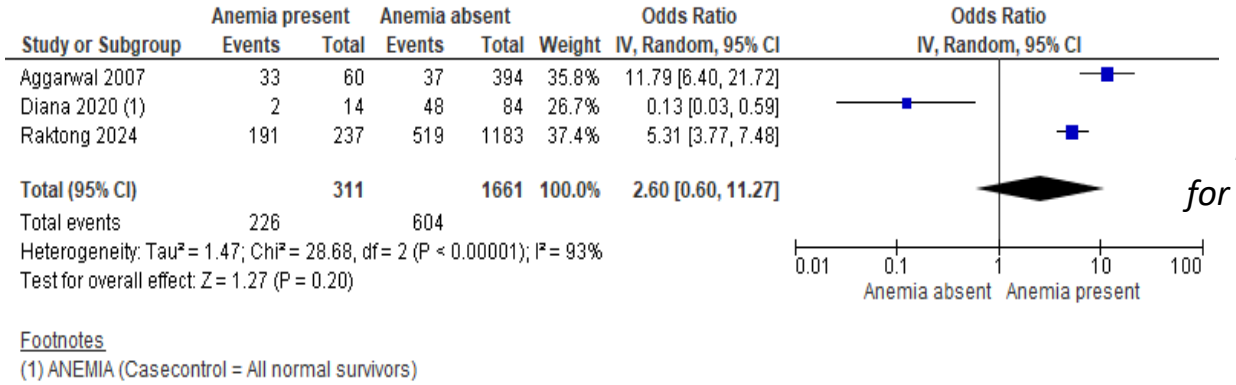
Subgroup analysis

Meta-regression

Sensitivity analysis

- A series of analyses **(re-analysis)** of a data set to assess whether **altering any of the assumptions made** leads to different final interpretations or conclusions
- Addresses the *"what-if-the-key-inputs-or-assumptions-changed"*-type of question
- **Exclusion of some studies -** sample size, methodological quality etc
- Different **meta-analysis models**
- Play a crucial role in **assessing the robustness of the findings** or conclusions

Association of anaemia complicating pregnancy, maternal age and parity with maternal mortality in the WHO South-East Asian region: a Systematic Review and Meta-analysis of observational studies

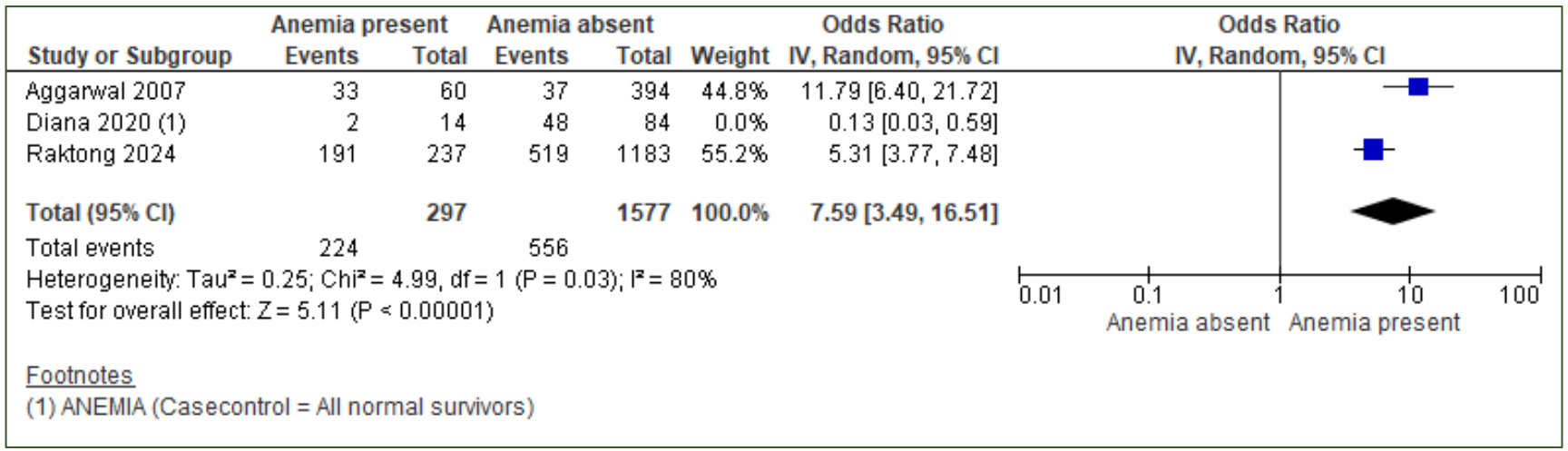


Study or Subgroup	Anemia present Events	Total	Anemia absent Events	Total	Weight	Odds Ratio IV, Random, 95% CI
Aggarwal 2007	33	60	37	394	35.8%	11.79 [6.40, 21.72]
Diana 2020 (1)	2	14	48	84	26.7%	0.13 [0.03, 0.59]
Raktong 2024	191	237	519	1183	37.4%	5.31 [3.77, 7.48]
Total (95% CI)		**311**		**1661**	**100.0%**	**2.60 [0.60, 11.27]**
Total events	226		604			

Heterogeneity: Tau² = 1.47; Chi² = 28.68, df = 2 (P < 0.00001); I² = 93%
Test for overall effect: Z = 1.27 (P = 0.20)

Footnotes
(1) ANEMIA (Casecontrol = All normal survivors)

Forest-plot showing the association of anemia status and maternal mortality for case-control studies



Study or Subgroup	Anemia present Events	Total	Anemia absent Events	Total	Weight	Odds Ratio IV, Random, 95% CI
Aggarwal 2007	33	60	37	394	44.8%	11.79 [6.40, 21.72]
Diana 2020 (1)	2	14	48	84	0.0%	0.13 [0.03, 0.59]
Raktong 2024	191	237	519	1183	55.2%	5.31 [3.77, 7.48]
Total (95% CI)		**297**		**1577**	**100.0%**	**7.59 [3.49, 16.51]**
Total events	224		556			

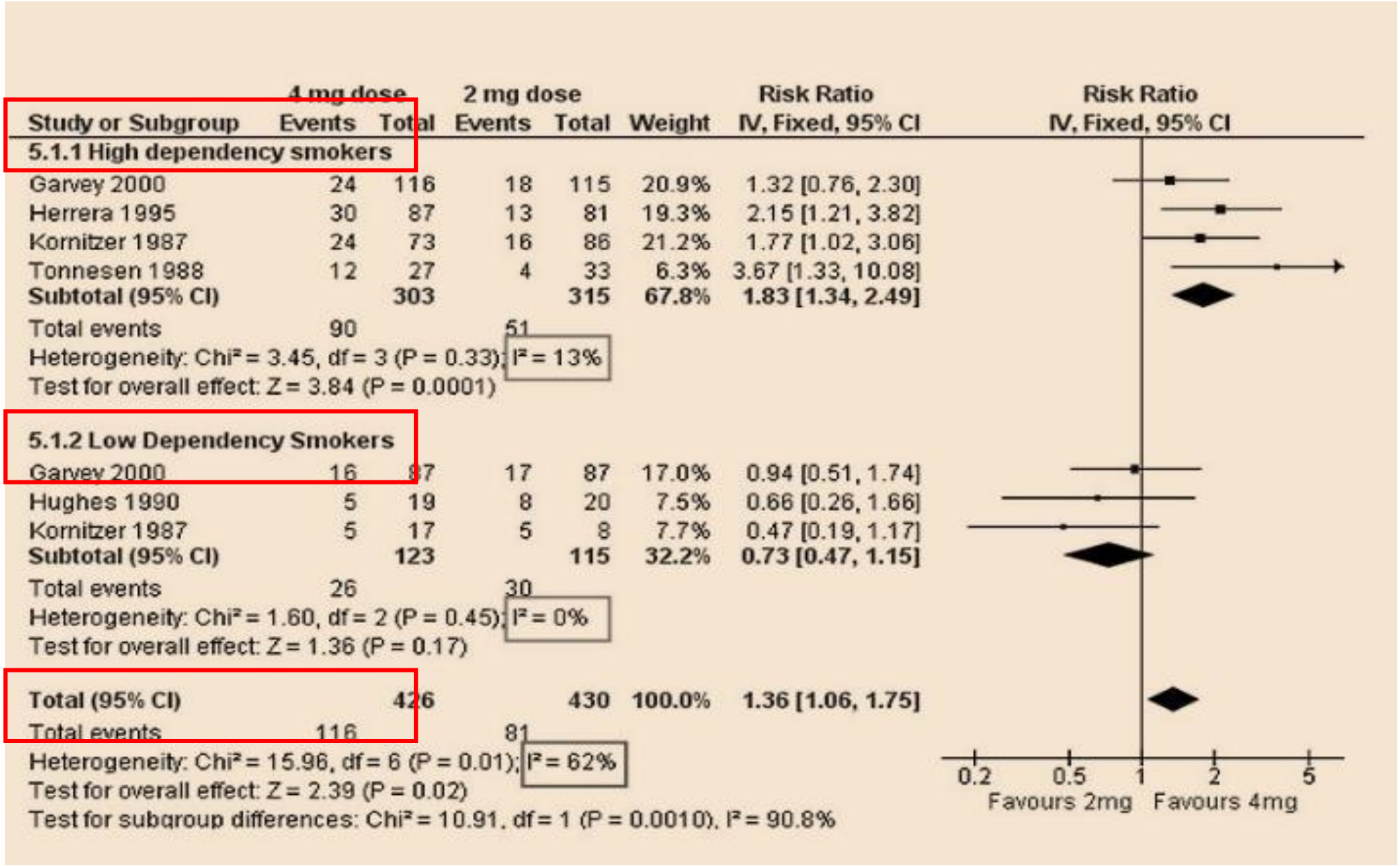
Heterogeneity: Tau² = 0.25; Chi² = 4.99, df = 1 (P = 0.03); I² = 80%
Test for overall effect: Z = 5.11 (P < 0.00001)

Footnotes
(1) ANEMIA (Casecontrol = All normal survivors)

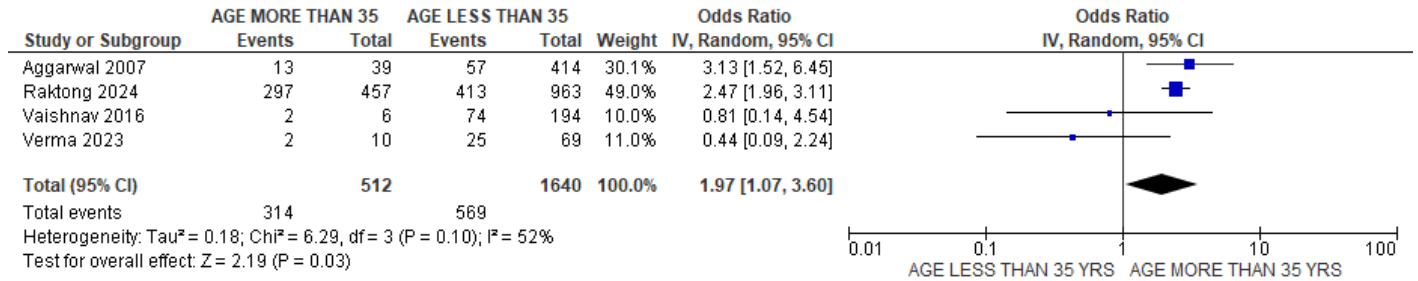
*Forest-plot showing the association of anemia status and maternal mortality for case-control studies on **sensitivity analysis** by removing poor quality study(Diana et al)*

Subgroup analysis

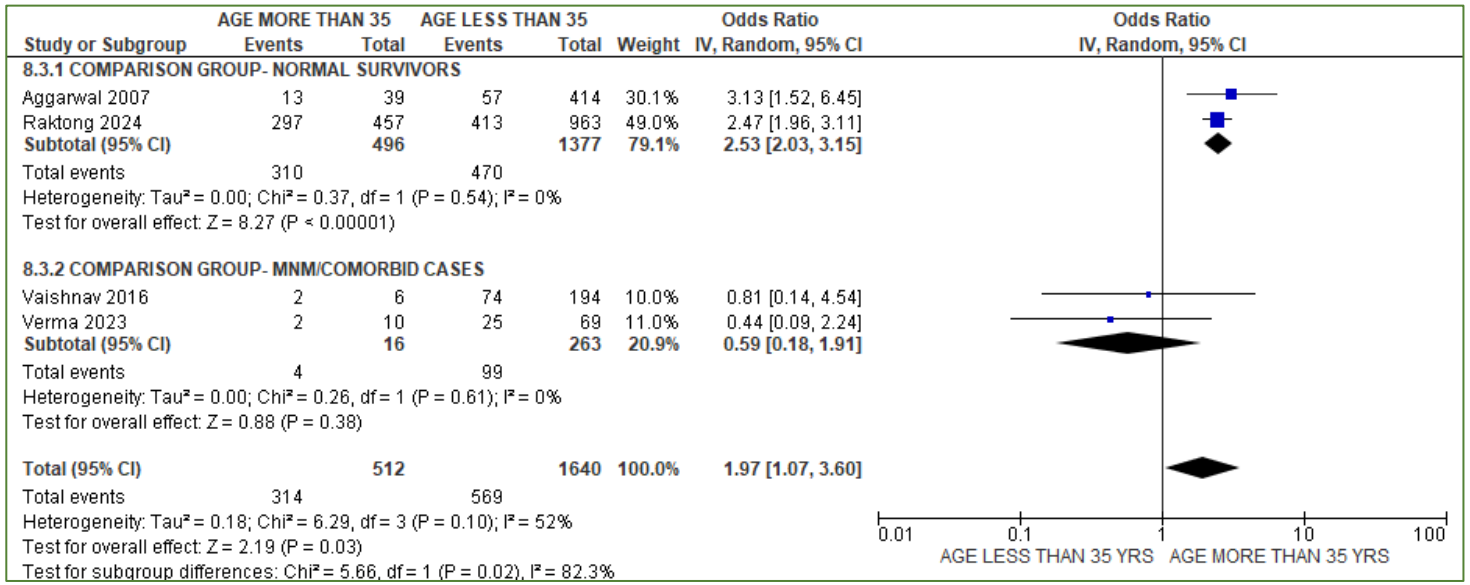
- Involve **splitting the data into subgroups** and comparing them
- **Done for** subsets - gender, age group, location , treatment regimen, designs
- Possible only for known characteristics of the populations
- To be **decided a priori**
- Main aim : to identify **consistency of the magnitude of treatment effect** among different categories of patients


4 mg dose
2 mg dose
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight IV, Fixed, 95% CI
IV, Fixed, 95% CI
5.1.1 High dependency smokers
Garvey 2000 24 116 18 115 20.9% 1.32 [0.76, 2.30]
Herrera 1995 30 87 13 81 19.3% 2.15 [1.21, 3.82]
Kornitzer 1987 24 73 16 86 21.2% 1.77 [1.02, 3.06]
Tonnesen 1988 12 27 4 33 6.3% 3.67 [1.33, 10.08]
Subtotal (95% CI) 303 315 67.8% 1.83 [1.34, 2.49]
Total events 90 51
Heterogeneity: Chi² = 3.45, df = 3 (P = 0.33); I² = 13%
Test for overall effect: Z = 3.84 (P = 0.0001)
5.1.2 Low Dependency Smokers
Garvey 2000 16 87 17 87 17.0% 0.94 [0.51, 1.74]
Hughes 1990 5 19 8 20 7.5% 0.66 [0.26, 1.66]
Kornitzer 1987 5 17 5 8 7.7% 0.47 [0.19, 1.17]
Subtotal (95% CI) 123 115 32.2% 0.73 [0.47, 1.15]
Total events 26 30
Heterogeneity: Chi² = 1.60, df = 2 (P = 0.45); I² = 0%
Test for overall effect: Z = 1.36 (P = 0.17)
Total (95% CI) 426 430 100.0% 1.36 [1.06, 1.75]
Total events 116 81
Heterogeneity: Chi² = 15.96, df = 6 (P = 0.01); I² = 62%
Test for overall effect: Z = 2.39 (P = 0.02)
Test for subgroup differences: Chi² = 10.91, df = 1 (P = 0.0010), I² = 90.8%
0.2
0.5
1
2
5
Favours 2mg
Favours 4mg

Association of anaemia complicating pregnancy, maternal age and parity with maternal mortality in the WHO South-East Asian region: a Systematic Review and Meta-analysis of observational studies

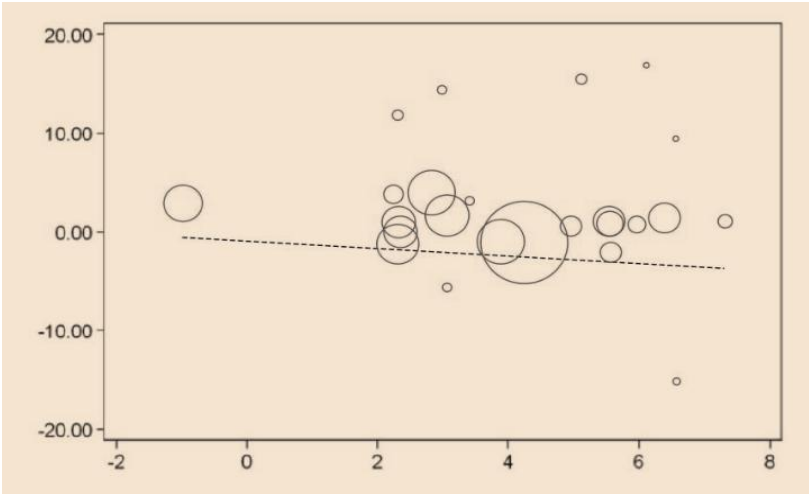


Forest plot showing the association of age more than 35 years and maternal mortality for case-control studies



*Forest-plot showing **subgroup analysis** of case-control studies showing the association of age more than 35 years and maternal mortality*

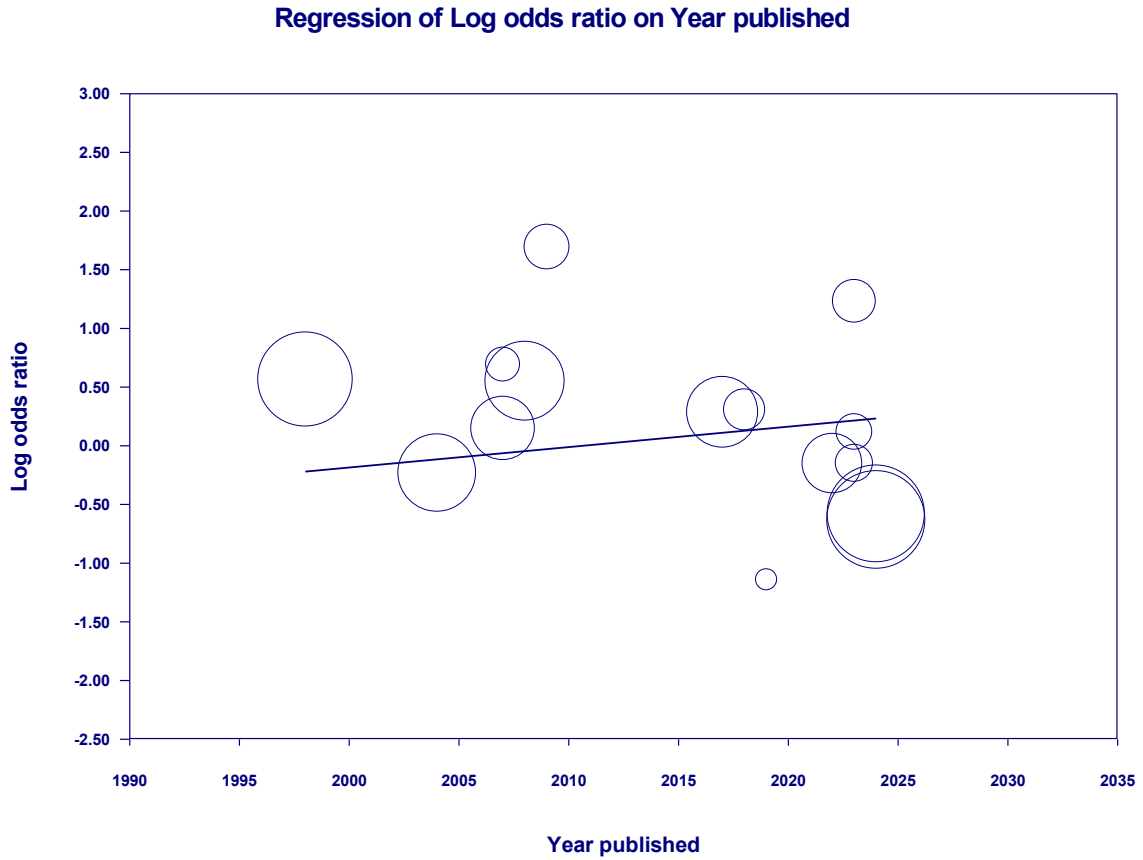
Meta-regression



- A modeling approach similar to multivariable analysis
- **Used to explore reasons for heterogeneity**
- **Determine the influence of selected variables (the independent variables) on the effect size (the dependent variable)**
  - **Dependent variable- Outcomes** - reported effect sizes
  - **Independent variables** - different age group, gender, location of study, year of study
  - Number of observations - **number of studies** in the meta-analysis

Meta-regression for the log odds ratio of primigravida (I^2 – 86%)

Covariates included in the model : 1. Year published (continuous variable) 2.Study Setting: Hospital based Vs Community Based 3. Type of Comparison group- MNM/Co-morbid survivors vs Normal survivors, 4. Design Cohort Vs Others 5. Country: India Vs Others



Bubble plot showing the effect of Primigravida on Maternal Mortality on Meta-regression

Covariate	β Coefficient	Standard Error	95% Lower	95% Upper	Z-value	2-sided P-value
Intercept	-34.0867	38.4074	-109.3638	41.1905	-0.89	0.3748
Year published	0.0174	0.0192	-0.0202	0.055	0.91	0.3638
Study Setting: Hospital based Vs Community based	-1.5376	0.4702	-2.4592	-0.6159	-3.27	**0.0011**
Comparison group type- MNM/Comorbid cases Vs Normal survivors	0.856	0.3204	0.2279	1.484	2.67	**0.0076**
Design: Cohort Vs Case-control	-0.241	0.2219	-0.6758	0.1939	-1.09	0.2775
Country: Others Vs India	-0.1789	0.2409	-0.651	0.2932	-0.74	0.4576

R^2 analog: 0.83

- Study setting and comparison group type significantly influence the relationship between primigravida and maternal mortality.

Bias in meta-analysis

- **Selection bias**
    - Inclusion criteria bias
    - Selector bias
- **Reporting bias**

TYPE OF REPORTING BIAS	DEFINITION
Publication bias	Publication and non-publication of research findings
Time lag bias	Delayed publication of some research findings
Multiple publication bias	Multiple publication of research findings
Citation bias	Citation or non-citation of certain research findings
Language bias	Publication of research findings in a particular language
Outcome reporting bias	Selective reporting of some findings only

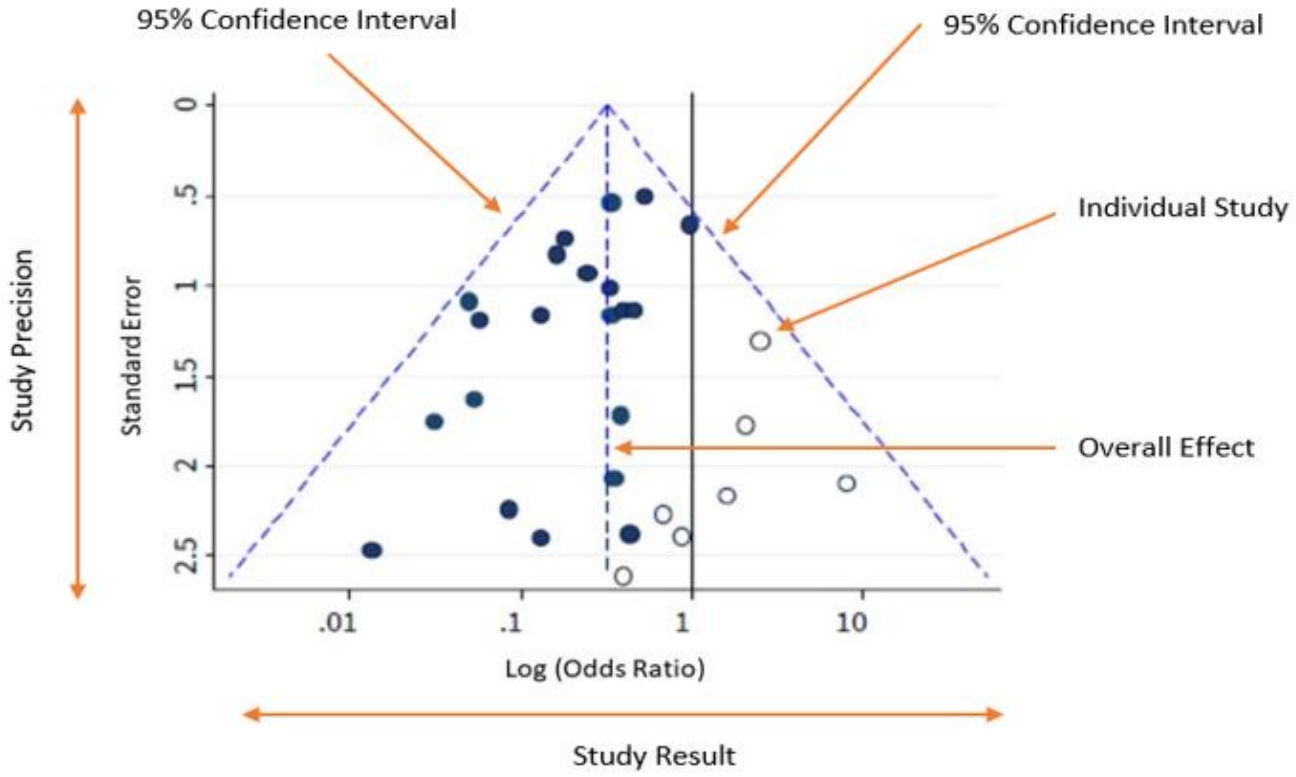
Publication bias



- Publication and non-publication of research findings
- Sources of publication bias :
    - Author
    - Sponsor of the trial
    - Editorial policy of the journal to which the paper has been submitted
    - Language
    - Popularity of research area
- Funnel plot

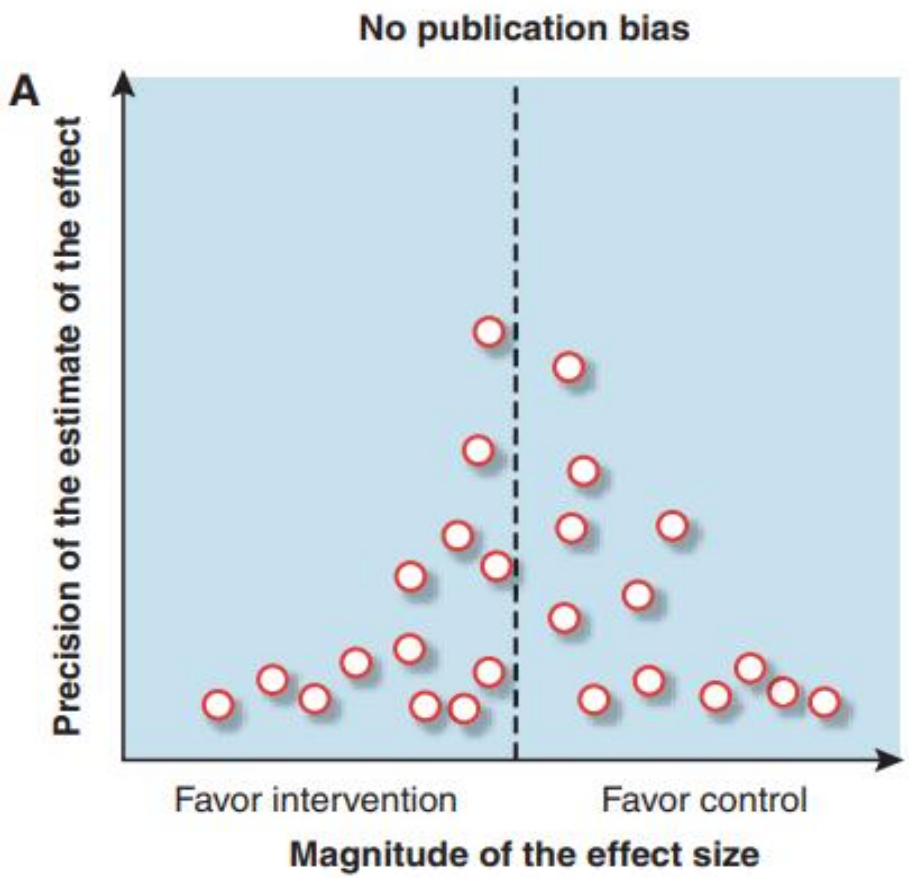
Funnel plot

- Scatter diagram
- Results of small studies will scatter widely
- Symmetry –no publication bias

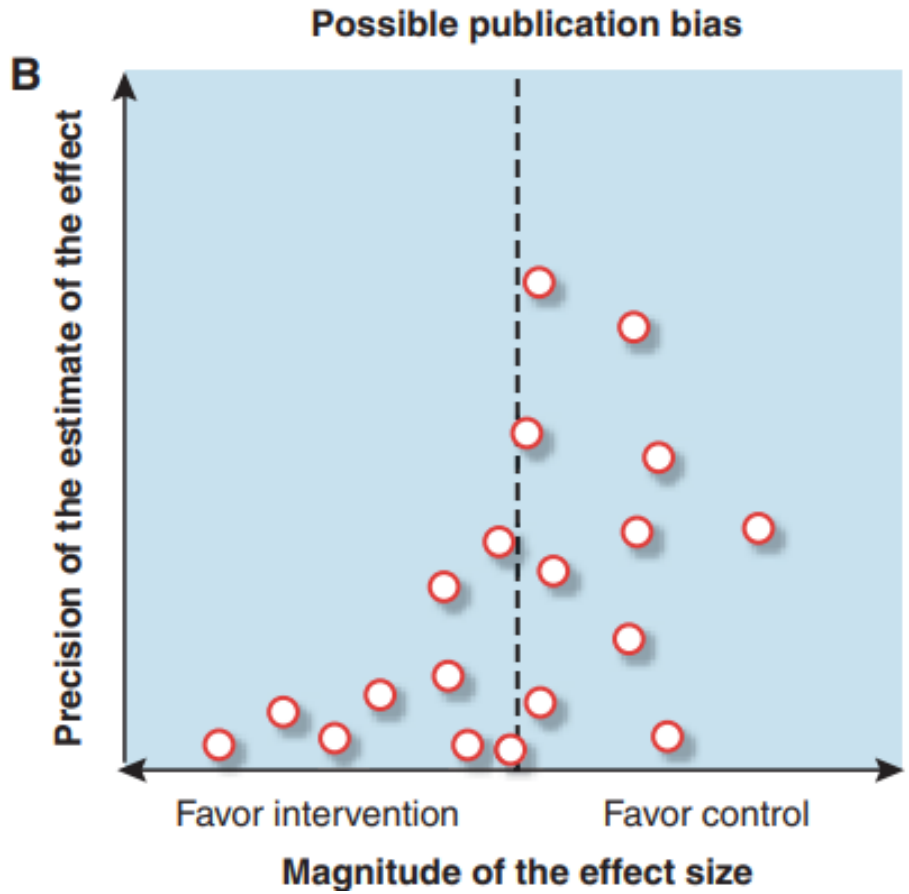


X -axis - Effect size

Y –axis - Precision (Sample size or Std error in reverse scale)

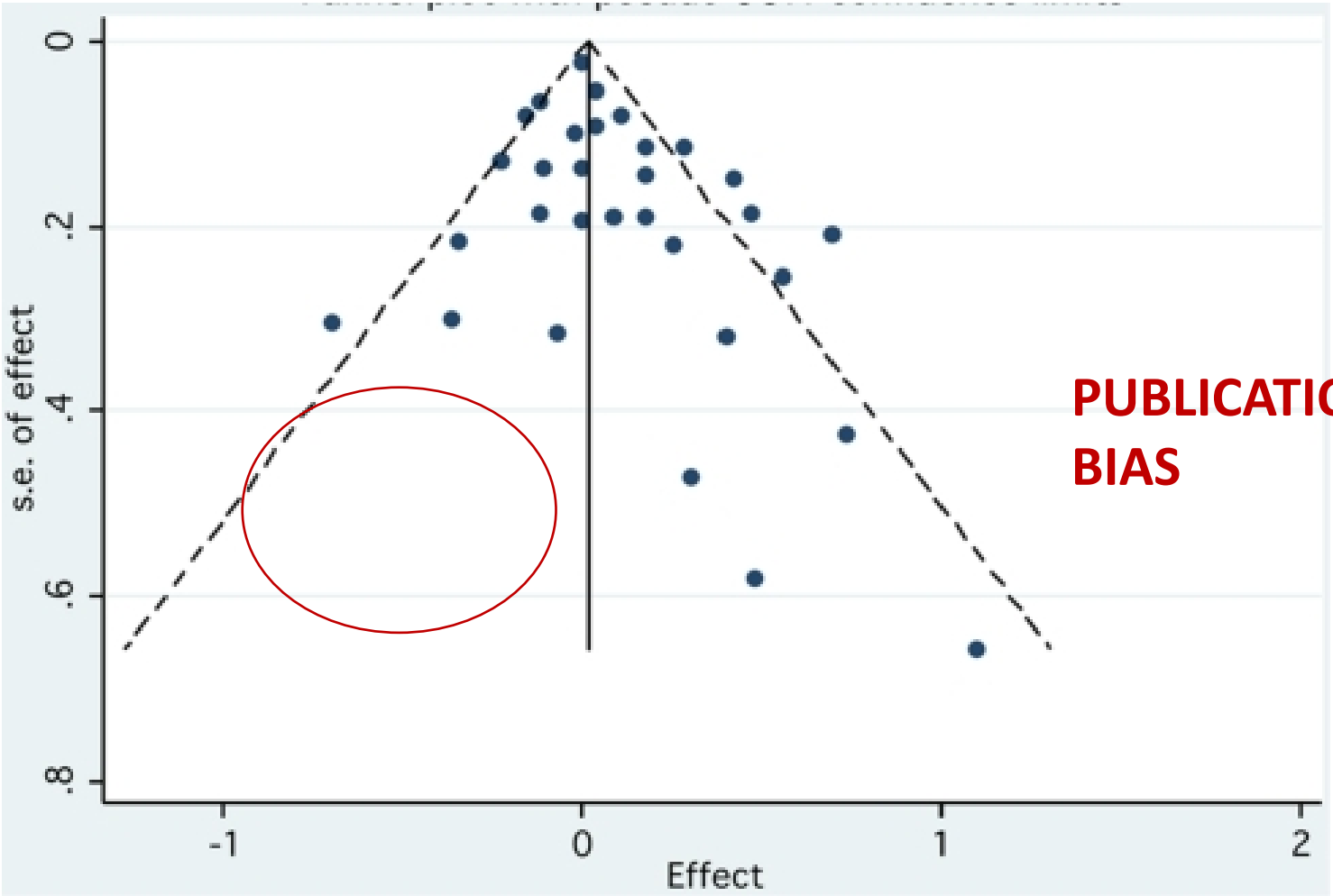


Trials are symmetrical in the shape of an inverted funnel, **suggesting no publication bias**



There are no trials in the lower right corner, suggesting that **small trials not favoring the intervention were not published.**

…Funnel plot



PUBLICATION BIAS

Disadvantage

- Subjective (Visual)
- Unreliable for small meta analysis

Statistical tests for publication bias

Linear regression test by Egger

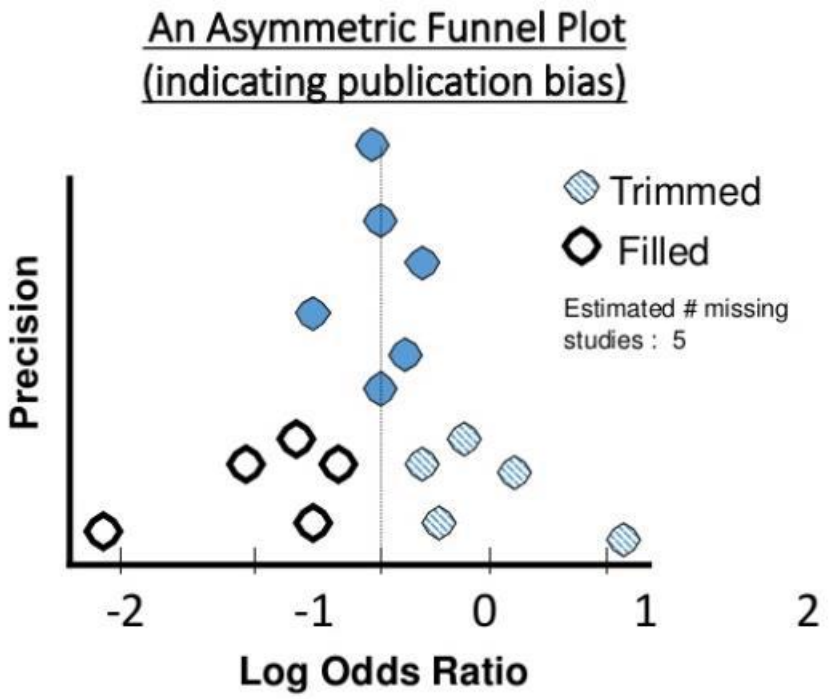
Begg's rank correlation coefficient

- Significant p-value: Suggests potential publication bias.
- Non-significant p-value: Does not necessarily rule out publication bias, as tests may have low power.

Adjusting for publication bias

Trim and fill method

Trim and Fill analysis adjusts for publication bias by identifying and incorporating hypothetical missing studies, **recalculating the combined effect size to provide a more accurate estimate**



Association of anaemia complicating pregnancy, maternal age and parity with maternal mortality in the WHO South-East Asian region: a Systematic Review and Meta-analysis of observational studies

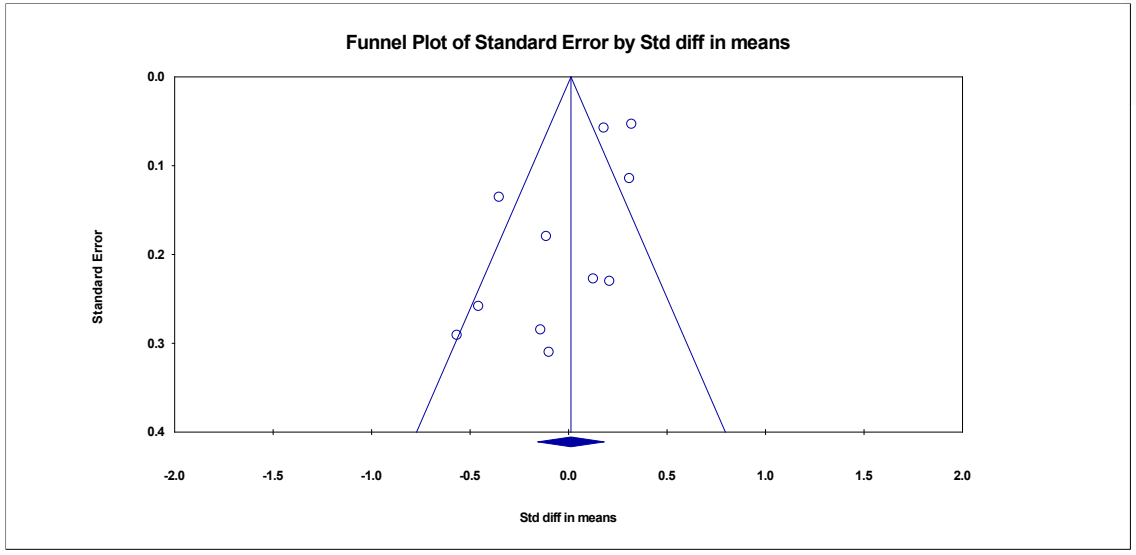


Figure showing funnel plot for the exposure variable , mean maternal age, where SMD is plotted

Duval and Tweedie's Trim and Fill

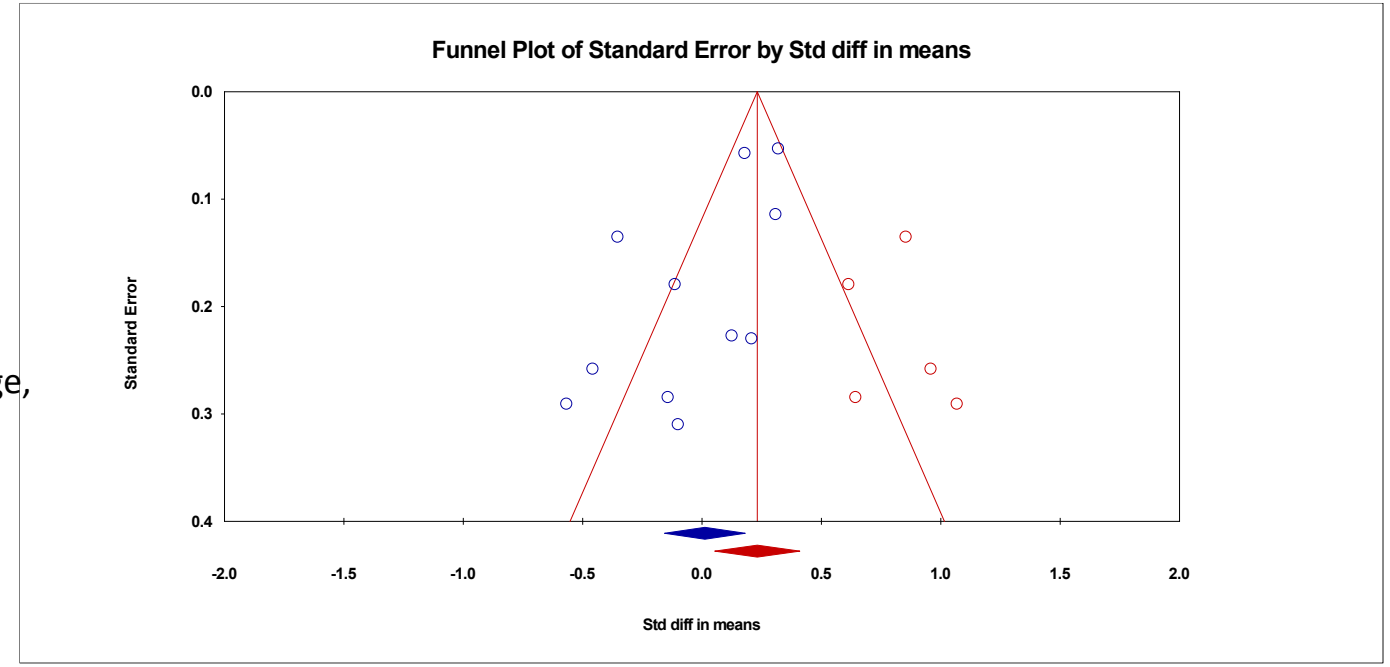


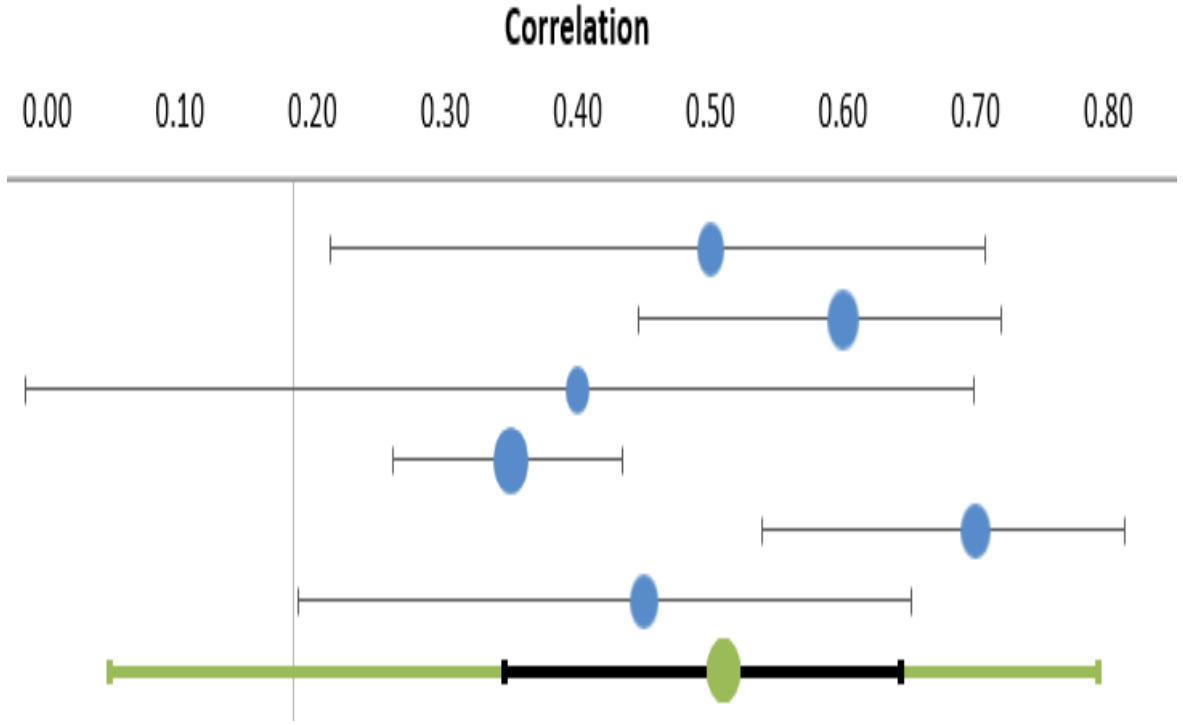
Figure showing funnel plot for the exposure variable , mean maternal age, where SMD is plotted using Trim and Fill method

- Observed SMD was 0.01 (95% CI: -0.15, 0.18)
- Adjusted SMD was found out to be 0.23 (95% CI: 0.05,0.40).

Confidence Interval vs Prediction Interval

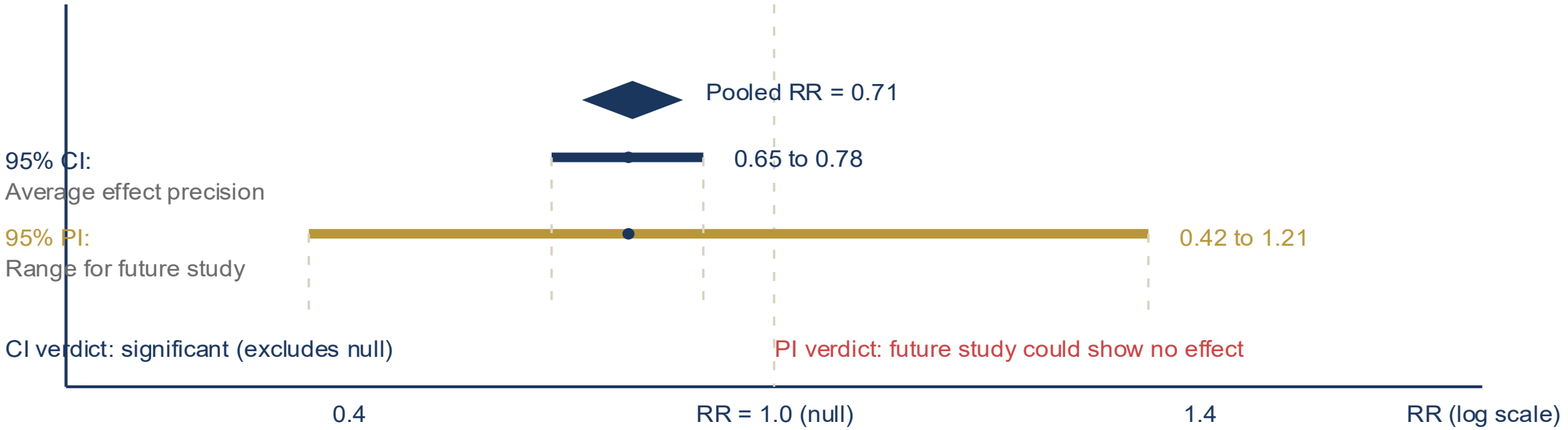
Confidence Interval vs Prediction Interval

- The confidence interval quantifies the accuracy of the mean, i.e. 95% of cases the mean effect size falls inside the horizontal black lines.
- Prediction interval addresses the actual dispersion of effect sizes, i.e. 95% of cases the true effect in a new study will fall inside the horizontal green lines



Prediction interval vs confidence interval

Both come from the same model. They answer different questions and are interpreted differently.



"With 95% confidence, the average effect across studies in this meta-analysis lies between these two values."

Width depends on number of studies and within-study precision.

"In a new study from a similar population, the true effect is expected to fall between these two values 95% of the time."

Width depends on tau-squared (between-study heterogeneity).

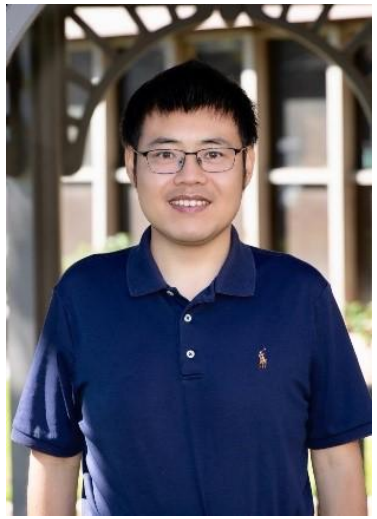
Reporting of analysis results- Reporting Guidelines

- **PRISMA**-Preferred Reporting Items for Systematic Reviews and Meta-Analysis
- **MOOSE**-Meta-analysis Of Observational Studies in Epidemiology
- **QUOROM** –Quality Control Of Reporting Of Meta-analysis

Meta-analysis Softwares

SOFTWARE	Free/ Paid	Developers
RevMan	Free	Cochrane
STATA	Paid	STATA Corp
SPSS	Paid	SPSS Inc
R	Free	CRAN
CMA	Paid	CMA Corp
MedCalc	Paid	MedCalc Inc
Mix 2.0	Paid	Biostat XL
OpenMeta Analyst	Free	CEBM
MAStARI	Free	JBI

Team for AI-Empowered Smart Healthy Ageing





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Qiu Huang
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Email: qhuang@mpu.edu.mo

Hsuan Yu Lu, James
Lecturer
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